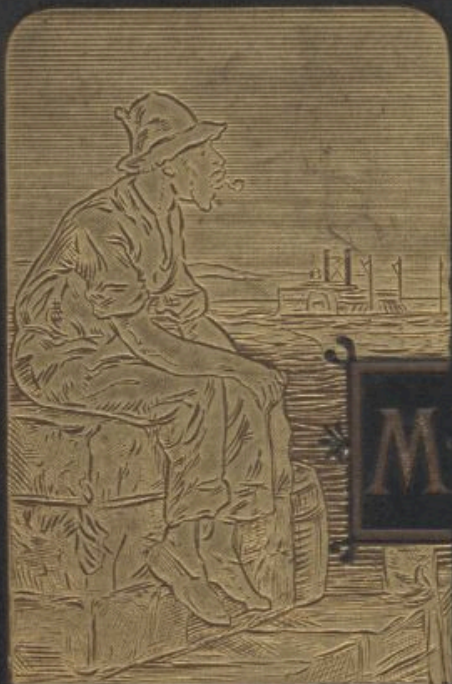




LIFE
ON THE
MISSISSIPPI

MARK TWAIN

ILLUSTRATED



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LIFE ON THE MISSISSIPPI, Part 2.

BY MARK TWAIN

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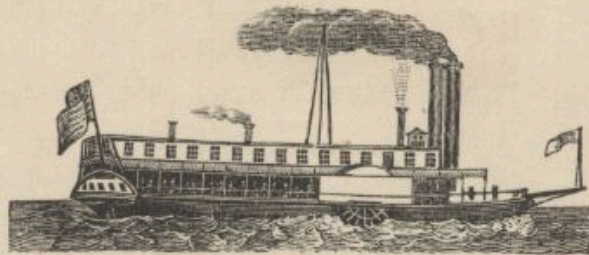
LIFE ON THE MISSISSIPPI

BY

MARK TWAIN

AUTHOR OF "THE INNOCENTS ABROAD," "ROUGHING IT,"
"THE PRINCE AND THE PAUPER," ETC.

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Mississippi Steamboat of Fifty Years Ago.

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BOSTON

JAMES R. OSGOOD AND COMPANY

1883

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Chapter 6

A Cub-pilot's Experience

WHAT with lying on the rocks four days at Louisville, and some other delays, the poor old 'Paul Jones' fooled away about two weeks in making the voyage from Cincinnati to New Orleans. This gave me a chance to get acquainted with one of the pilots, and he taught me how to steer the boat, and thus made the fascination of river life more potent than ever for me.

It also gave me a chance to get acquainted with a youth who had taken deck passage—more's the pity; for he easily borrowed six dollars of me on a promise to return to the boat and pay it back to me the day after we should arrive. But he probably died or forgot, for he never came. It was doubtless the former, since he had said his parents were wealthy, and he only traveled deck passage because it was cooler.{footnote [1. 'Deck' Passage, i.e. steerage passage.]}



I soon discovered two things. One was that a vessel would not be likely to sail for the mouth of the Amazon under ten or twelve years; and the other was that the nine or ten dollars still left in my pocket would not suffice for so imposing an exploration as I had planned, even if I could afford to wait for a ship. Therefore it followed that I must contrive a new career. The 'Paul Jones' was now bound for St. Louis. I planned a siege against my pilot, and at the end of three hard days he surrendered. He agreed to teach me the Mississippi River from New Orleans to St. Louis for five hundred dollars, payable out of the first wages I should receive after graduating. I entered upon the small enterprise of 'learning' twelve or thirteen hundred miles of the great Mississippi River with the easy confidence of my time of life. If I

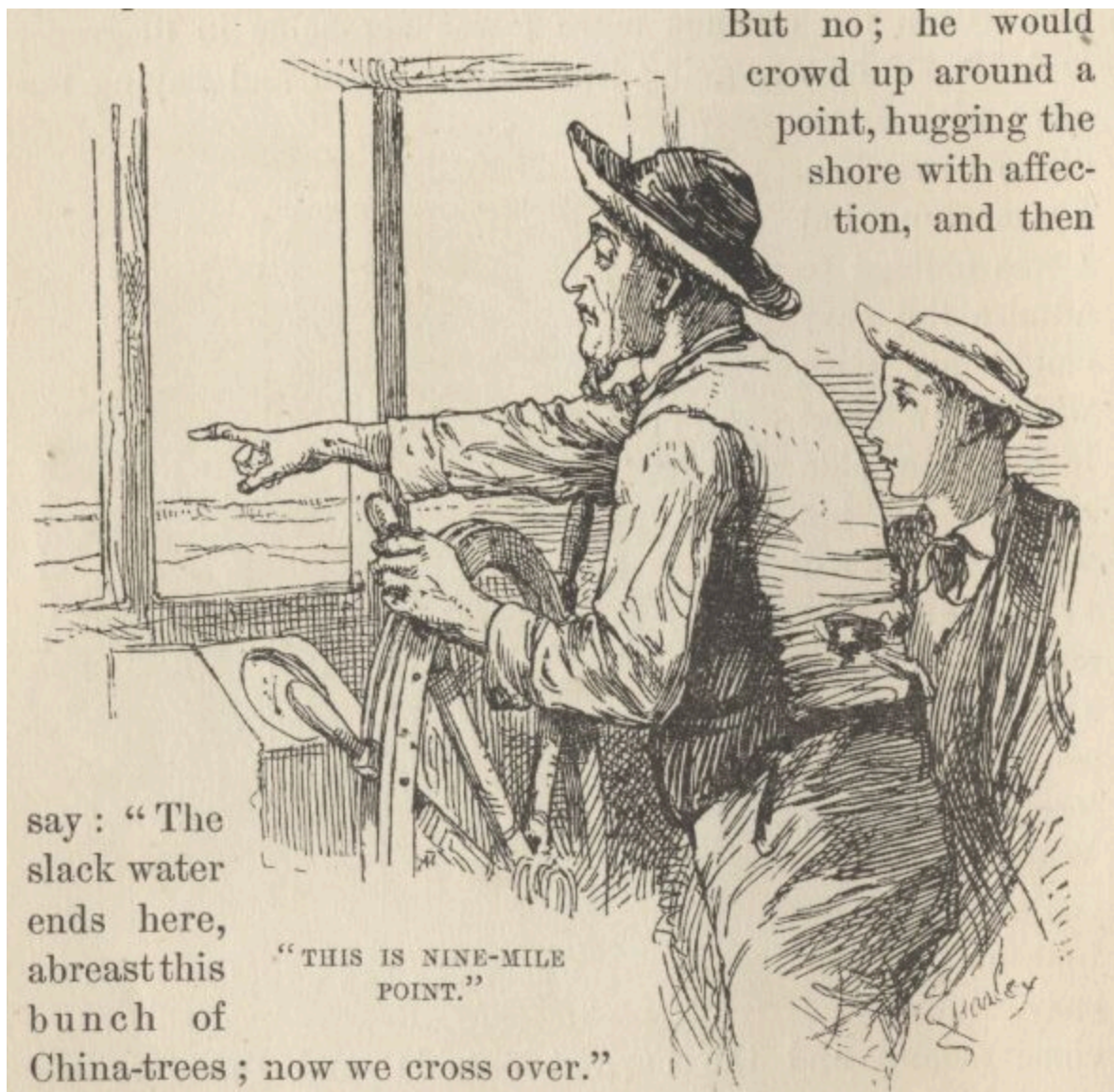
had really known what I was about to require of my faculties, I should not have had the courage to begin. I supposed that all a pilot had to do was to keep his boat in the river, and I did not consider that that could be much of a trick, since it was so wide.



The boat backed out from New Orleans at four in the afternoon, and it was 'our watch' until eight. Mr. Bixby, my chief, 'straightened her up,' plowed her along past the sterns of the other boats that lay at the Levee, and then said, 'Here, take her; shave those steamships as close as you'd peel an apple.' I took the wheel, and my heart-beat fluttered up into the hundreds; for it seemed to me that we were about to scrape the side off every ship in

the line, we were so close. I held my breath and began to claw the boat away from the danger; and I had my own opinion of the pilot who had known no better than to get us into such peril, but I was too wise to express it. In half a minute I had a wide margin of safety intervening between the 'Paul Jones' and the ships; and within ten seconds more I was set aside in disgrace, and Mr. Bixby was going into danger again and flaying me alive with abuse of my cowardice. I was stung, but I was obliged to admire the easy confidence with which my chief loafed from side to side of his wheel, and trimmed the ships so closely that disaster seemed ceaselessly imminent. When he had cooled a little he told me that the easy water was close ashore and the current outside, and therefore we must hug the bank, up-stream, to get the benefit of the former, and stay well out, down-stream, to take advantage of the latter. In my own mind I resolved to be a down-stream pilot and leave the up-streaming to people dead to prudence.

Now and then Mr. Bixby called my attention to certain things. Said he, 'This is Six-Mile Point.' I assented. It was pleasant enough information, but I could not see the bearing of it. I was not conscious that it was a matter of any interest to me. Another time he said, 'This is Nine-Mile Point.' Later he said, 'This is Twelve-Mile Point.' They were all about level with the water's edge; they all looked about alike to me; they were monotonously unpicturesque. I hoped Mr. Bixby would change the subject. But no; he would crowd up around a point, hugging the shore with affection, and then say: 'The slack water ends here, abreast this bunch of China-trees; now we cross over.' So he crossed over. He gave me the wheel once or twice, but I had no luck. I either came near chipping off the edge of a sugar plantation, or I yawed too far from shore, and so dropped back into disgrace again and got abused.



But no; he would crowd up around a point, hugging the shore with affection, and then

say : "The slack water ends here, abreast this bunch of China-trees ; now we cross over."

"THIS IS NINE-MILE POINT."

The watch was ended at last, and we took supper and went to bed. At midnight the glare of a lantern shone in my eyes, and the night watchman said—

'Come! turn out!'



And then he left. I could not understand this extraordinary procedure; so I presently gave up trying to, and dozed off to sleep. Pretty soon the watchman was back again, and this time he was gruff. I was annoyed. I said:—

'What do you want to come bothering around here in the middle of the night for. Now as like as not I'll not get to sleep again to-night.'

The watchman said—

'Well, if this an't good, I'm blest.'

The 'off-watch' was just turning in, and I heard some brutal laughter from them, and such remarks as 'Hello, watchman! an't the new cub turned out yet? He's delicate, likely. Give him some sugar in a rag and send for the chambermaid to sing rock-a-by-baby to him.'



"A MINUTE LATER."

About this time Mr. Bixby appeared on the scene. Something like a minute later I was climbing the pilot-house steps with some of my clothes on and the rest in my arms. Mr. Bixby was close behind, commenting. Here was something fresh—this thing of getting up in the middle of the night to go to work. It was a detail in piloting that had never occurred to me at all. I knew that boats ran all night, but somehow I had never happened to reflect that somebody had to get up out of a warm bed to run them. I began to fear that piloting was not quite so romantic as I had imagined it was; there was something very real and work-like about this new phase of it.

It was a rather dingy night, although a fair number of stars were out. The big mate was at the wheel, and he had the old tub pointed at a star and was holding her straight up the middle of the river. The shores on either hand were not much more than half a mile apart, but they seemed wonderfully far away and ever so vague and indistinct. The mate said:—

'We've got to land at Jones's plantation, sir.'

The vengeful spirit in me exulted. I said to myself, I wish you joy of your job, Mr. Bixby; you'll have a good time finding Mr. Jones's plantation such a night as this; and I hope you never WILL find it as long as you live.

Mr. Bixby said to the mate:—

'Upper end of the plantation, or the lower?'

'Upper.'

'I can't do it. The stumps there are out of water at this stage: It's no great distance to the lower, and you'll have to get along with that.'

'All right, sir. If Jones don't like it he'll have to lump it, I reckon.'

And then the mate left. My exultation began to cool and my wonder to come up. Here was a man who not only proposed to find this plantation on such a night, but to find either end of it you preferred. I dreadfully wanted to ask a question, but I was carrying about as many short answers as my cargo-room would admit of, so I held my peace. All I desired to ask Mr. Bixby was the simple question whether he was ass enough to really imagine he was going to find that plantation on a night when all plantations were

exactly alike and all the same color. But I held in. I used to have fine inspirations of prudence in those days.

Mr. Bixby made for the shore and soon was scraping it, just the same as if it had been daylight. And not only that, but singing—

'Father in heaven, the day is declining,' etc.

It seemed to me that I had put my life in the keeping of a peculiarly reckless outcast. Presently he turned on me and said:—

'What's the name of the first point above New Orleans?'

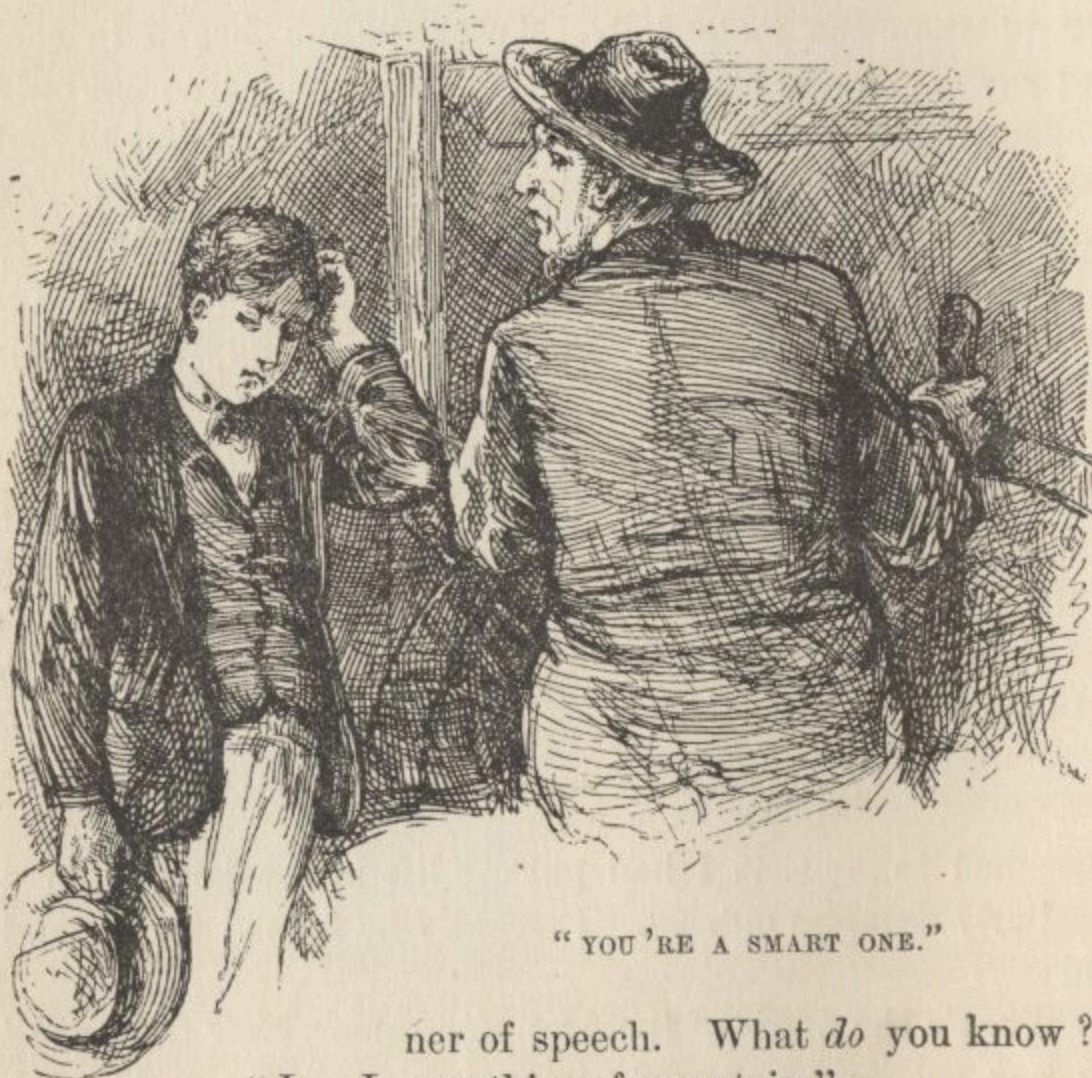
I was gratified to be able to answer promptly, and I did. I said I didn't know.

'Don't KNOW?'

This manner jolted me. I was down at the foot again, in a moment. But I had to say just what I had said before.

'Well, you're a smart one,' said Mr. Bixby. 'What's the name of the NEXT point?'

"You—you—don't know?" mimicking my drawling man-



"YOU'RE A SMART ONE."

ner of speech. What *do* you know?"

"I—I—nothing, for certain."

"By the great Cæsar's ghost, I believe you! You're the stupidest dunderhead I ever saw or ever heard of, so help me Moses! The idea of *you* being a pilot—*you*! Why, you don't know enough to pilot a cow down a lane."

Once more I didn't know.

'Well, this beats anything. Tell me the name of ANY point or place I told you.'

I studied a while and decided that I couldn't.

'Look here! What do you start out from, above Twelve-Mile Point, to cross over?'

'I—I—don't know.'

'You—you—don't know?' mimicking my drawling manner of speech. 'What DO you know?'

'I—I—nothing, for certain.'

'By the great Caesar's ghost, I believe you! You're the stupidest dunderhead I ever saw or ever heard of, so help me Moses! The idea of you being a pilot—you! Why, you don't know enough to pilot a cow down a lane.'

Oh, but his wrath was up! He was a nervous man, and he shuffled from one side of his wheel to the other as if the floor was hot. He would boil a while to himself, and then overflow and scald me again.

'Look here! What do you suppose I told you the names of those points for?'

I tremblingly considered a moment, and then the devil of temptation provoked me to say:—

'Well—to—to—be entertaining, I thought.'

This was a red rag to the bull. He raged and stormed so (he was crossing the river at the time) that I judge it made him blind, because he ran over the steering-oar of a trading-scow. Of course the traders sent up a volley of red-hot profanity. Never was a man so grateful as Mr. Bixby was: because he was brim full, and here were subjects who would TALK BACK. He threw open a window, thrust his head out, and such an irruption followed as I never had heard before. The fainter and farther away the scowmen's curses drifted, the higher Mr. Bixby lifted his voice and the weightier his adjectives grew. When he closed the window he was empty. You could have

drawn a seine through his system and not caught curses enough to disturb your mother with. Presently he said to me in the gentlest way—

'My boy, you must get a little memorandum book, and every time I tell you a thing, put it down right away. There's only one way to be a pilot, and that is to get this entire river by heart. You have to know it just like A B C.'

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"GET A MEMORANDUM
BOOK."

That was a dismal revelation to me; for my memory was never loaded with anything but blank cartridges. However, I did not feel discouraged long. I judged that it was best to make some allowances, for doubtless Mr. Bixby was 'stretching.' Presently he pulled a rope and struck a few strokes on the big bell. The stars were all gone now, and the night was as black as ink. I could hear the wheels churn along the bank, but I was not entirely certain that I could see the shore. The voice of the invisible watchman called up from the hurricane deck—

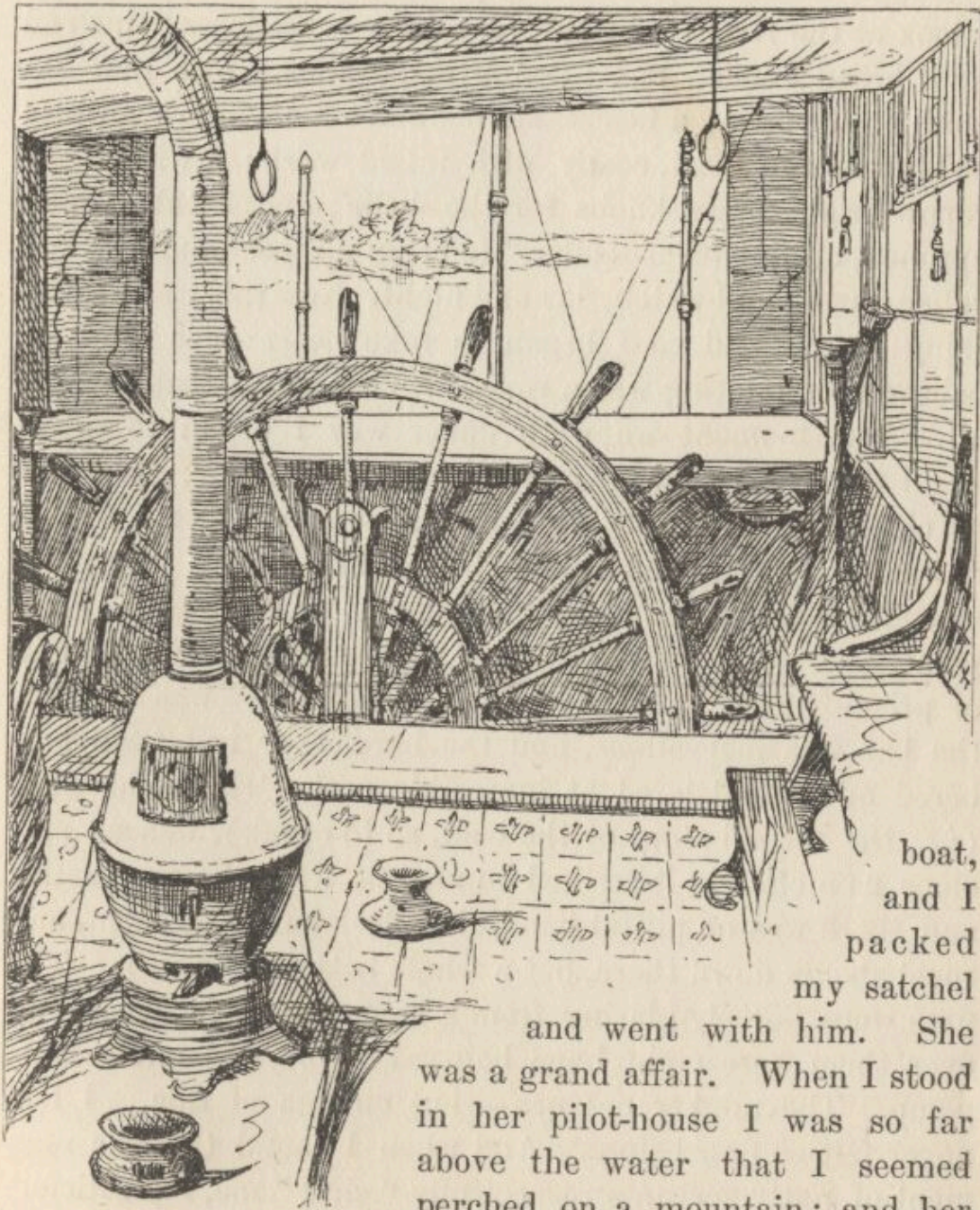
'What's this, sir?'

'Jones's plantation.'

I said to myself, I wish I might venture to offer a small bet that it isn't. But I did not chirp. I only waited to see. Mr. Bixby handled the engine bells, and in due time the boat's nose came to the land, a torch glowed from the forecastle, a man skipped ashore, a darky's voice on the bank said, 'Gimme de k'yarpet-bag, Mars' Jones,' and the next moment we were standing up the river again, all serene. I reflected deeply awhile, and then said—but not aloud—'Well, the finding of that plantation was the luckiest accident that ever happened; but it couldn't happen again in a hundred years.' And I fully believed it was an accident, too.

By the time we had gone seven or eight hundred miles up the river, I had learned to be a tolerably plucky up-stream steersman, in daylight, and before we reached St. Louis I had made a trifle of progress in night-work, but only a trifle. I had a note-book that fairly bristled with the names of towns, 'points,' bars, islands, bends, reaches, etc.; but the information was to be found only in the notebook—none of it was in my head. It made my heart ache to think I had only got half of the river set down; for as our watch was four hours off and four hours on, day and night, there was a long four-hour gap in my book for every time I had slept since the voyage began.

My chief was presently hired to go on a big New Orleans



"A SUMPTUOUS TEMPLE."

how I could ever have considered the little "Paul Jones" a large craft. There were other differences, too. The "Paul Jones's" pilot-house was a cheap, dingy, battered rattle-trap,

boat,
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was a grand affair. When I stood
in her pilot-house I was so far
above the water that I seemed
perched on a mountain; and her
decks stretched so far away, fore
and aft, below me, that I wondered

My chief was presently hired to go on a big New Orleans boat, and I packed my satchel and went with him. She was a grand affair. When I stood in her pilot-house I was so far above the water that I seemed perched on a mountain; and her decks stretched so far away, fore and aft, below me, that I wondered how I could ever have considered the little 'Paul Jones' a large craft. There were other differences, too. The 'Paul Jones's' pilot-house was a cheap, dingy, battered rattle-trap, cramped for room: but here was a sumptuous glass temple; room enough to have a dance in; showy red and gold window-curtains; an imposing sofa; leather cushions and a back to the high bench where visiting pilots sit, to spin yarns and 'look at the river;' bright, fanciful 'cuspadores' instead of a broad wooden box filled with sawdust; nice new oil-cloth on the floor; a hospitable big stove for winter; a wheel as high as my head, costly with inlaid work; a wire tiller-rope; bright brass knobs for the bells; and a tidy, white-aproned, black 'texas-tender,' to bring up tarts and ices and coffee during mid-watch, day and night. Now this was 'something like,' and so I began to take heart once more to believe that piloting was a romantic sort of occupation after all. The moment we were under way I began to prowls about the great steamer and fill myself with joy. She was as clean and as dainty as a drawing-room; when I looked down her long, gilded saloon, it was like gazing through a splendid tunnel; she had an oil-picture, by some gifted sign-painter, on every stateroom door; she glittered with no end of prism-fringed chandeliers; the clerk's office was elegant, the bar was marvelous, and the bar-keeper had been barbered and upholstered at incredible cost. The boiler deck (i.e. the second story of the boat, so to speak) was as spacious as a church, it seemed to me; so with the fore-castle; and there was no pitiful handful of deckhands, firemen, and roustabouts down there, but a whole battalion of men. The fires were fiercely glaring from a long row of furnaces, and over them were eight huge boilers! This was unutterable pomp. The mighty engines—but enough of this. I had never felt so fine before. And when I found that the regiment of natty servants respectfully 'sir'd' me, my satisfaction was complete.

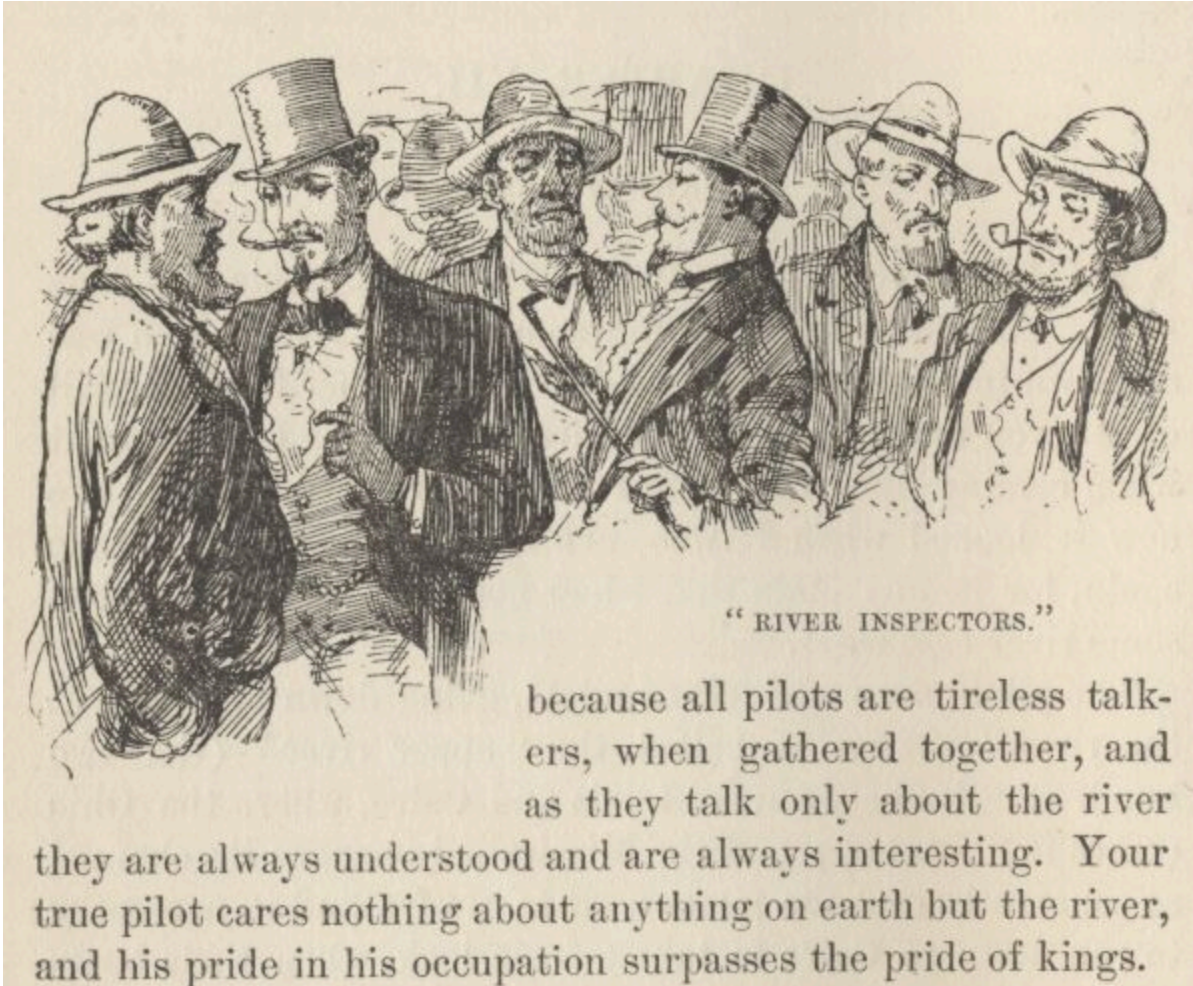
Chapter 7

A Daring Deed

WHEN I returned to the pilot-house St. Louis was gone and I was lost. Here was a piece of river which was all down in my book, but I could make neither head nor tail of it: you understand, it was turned around. I had seen it when coming up-stream, but I had never faced about to see how it looked when it was behind me. My heart broke again, for it was plain that I had got to learn this troublesome river BOTH WAYS.

The pilot-house was full of pilots, going down to 'look at the river.' What is called the 'upper river' (the two hundred miles between St. Louis and Cairo, where the Ohio comes in) was low; and the Mississippi changes its channel so constantly that the pilots used to always find it necessary to run down to Cairo to take a fresh look, when their boats were to lie in port a week; that is, when the water was at a low stage. A deal of this 'looking at the river' was done by poor fellows who seldom had a berth, and whose only hope of getting one lay in their being always freshly posted and therefore ready to drop into the shoes of some reputable pilot, for a single trip, on account of such pilot's sudden illness, or some other necessity. And a good many of them constantly ran up and down inspecting the river, not because they ever really hoped to get a berth, but because (they being guests of the boat) it was cheaper to 'look at the river' than stay ashore and pay board. In time these fellows grew dainty in their tastes, and only infested boats that had an established reputation for setting good tables. All visiting pilots were useful, for they were always ready and willing, winter or summer, night or day, to go out in the yawl and help buoy the channel or assist the boat's pilots in any way they could. They were likewise welcome because all pilots are tireless talkers, when gathered together, and as they talk only about the river they are always understood and are always

interesting. Your true pilot cares nothing about anything on earth but the river, and his pride in his occupation surpasses the pride of kings.



We had a fine company of these river-inspectors along, this trip. There were eight or ten; and there was abundance of room for them in our great pilot-house. Two or three of them wore polished silk hats, elaborate shirt-fronts, diamond breast-pins, kid gloves, and patent-leather boots. They were choice in their English, and bore themselves with a dignity proper to men of solid means and prodigious reputation as pilots. The others were more or less loosely clad, and wore upon their heads tall felt cones that were suggestive of the days of the Commonwealth.

I was a cipher in this august company, and felt subdued, not to say torpid. I was not even of sufficient consequence to assist at the wheel when it was necessary to put the tiller hard down in a hurry; the guest that stood nearest did that when occasion required—and this was pretty much all the time, because of the crookedness of the channel and the scant water. I stood in a corner; and the talk I listened to took the hope all out of me. One visitor said to another—

'Jim, how did you run Plum Point, coming up?'

'It was in the night, there, and I ran it the way one of the boys on the "Diana" told me; started out about fifty yards above the wood pile on the false point, and held on the cabin under Plum Point till I raised the reef—quarter less twain—then straightened up for the middle bar till I got well abreast the old one-limbed cotton-wood in the bend, then got my stern on the cotton-wood and head on the low place above the point, and came through a-booming—nine and a half.'

'Pretty square crossing, an't it?'

'Yes, but the upper bar 's working down fast.'

Another pilot spoke up and said—

'I had better water than that, and ran it lower down; started out from the false point—mark twain—raised the second reef abreast the big snag in the bend, and had quarter less twain.'

One of the gorgeous ones remarked—

'I don't want to find fault with your leadsmen, but that's a good deal of water for Plum Point, it seems to me.'

There was an approving nod all around as this quiet snub dropped on the boaster and 'settled' him. And so they went on talk-talk-talking. Meantime, the thing that was running in my mind was, 'Now if my ears hear aright, I have not only to get the names of all the towns and islands and bends, and so on, by heart, but I must even get up a warm personal acquaintanceship with every old snag and one-limbed cotton-wood and obscure wood pile that ornaments the banks of this river for twelve hundred miles; and more

than that, I must actually know where these things are in the dark, unless these guests are gifted with eyes that can pierce through two miles of solid blackness; I wish the piloting business was in Jericho and I had never thought of it.'

At dusk Mr. Bixby tapped the big bell three times (the signal to land), and the captain emerged from his drawing-room in the forward end of the texas, and looked up inquiringly. Mr. Bixby said—

'We will lay up here all night, captain.'

'Very well, sir.'

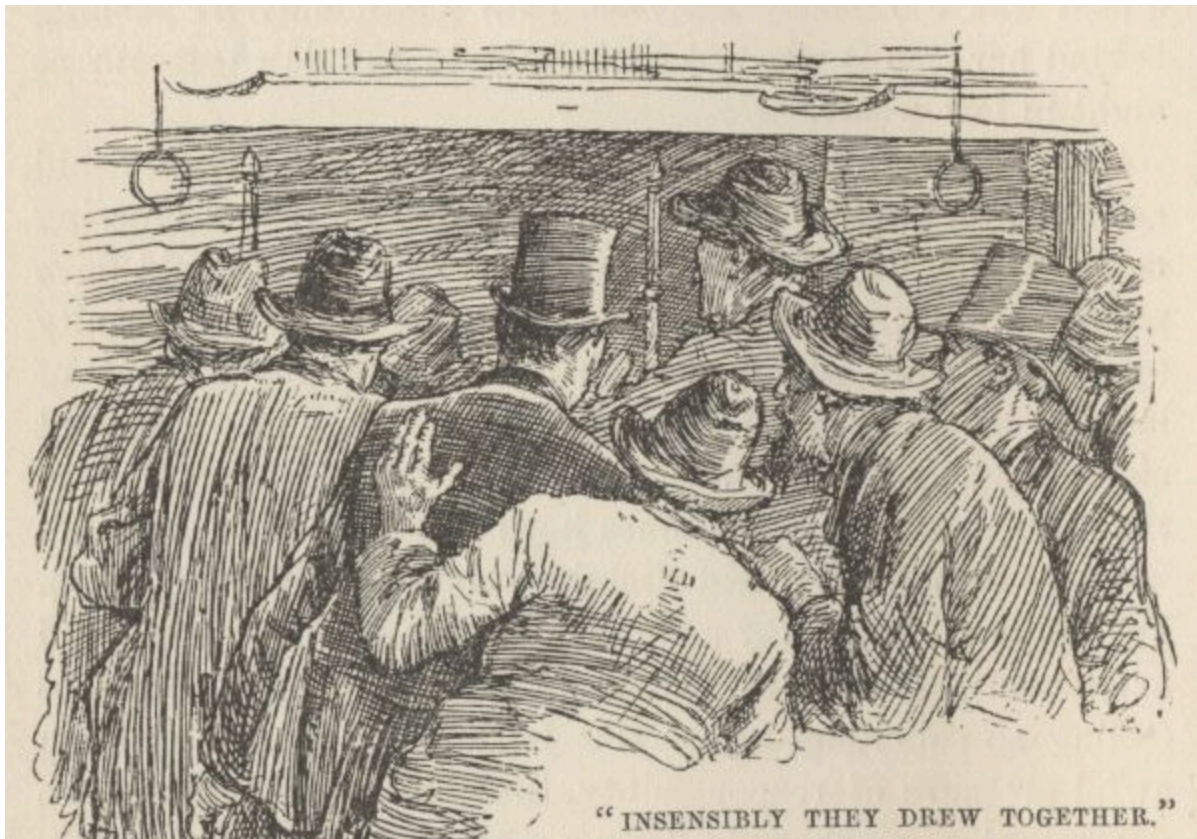
That was all. The boat came to shore and was tied up for the night. It seemed to me a fine thing that the pilot could do as he pleased, without asking so grand a captain's permission. I took my supper and went immediately to bed, discouraged by my day's observations and experiences. My late voyage's note-book was but a confusion of meaningless names. It had tangled me all up in a knot every time I had looked at it in the daytime. I now hoped for respite in sleep; but no, it reveled all through my head till sunrise again, a frantic and tireless nightmare.



Next morning I felt pretty rusty and low-spirited. We went booming along, taking a good many chances, for we were anxious to 'get out of the river' (as getting out to Cairo was called) before night should overtake us. But Mr. Bixby's partner, the other pilot, presently grounded the boat, and we lost so much time in getting her off that it was plain that darkness would overtake us a good long way above the mouth. This was a great misfortune, especially to certain of our visiting pilots, whose boats would have to wait for their return, no matter how long that might be. It sobered the pilot-house talk a good deal. Coming up-stream, pilots did not mind low water or any kind of darkness; nothing stopped them but fog. But down-stream work was different; a boat was too nearly helpless, with a stiff current pushing behind her; so it was not customary to run down-stream at night in low water.

There seemed to be one small hope, however: if we could get through the intricate and dangerous Hat Island crossing before night, we could venture the rest, for we would have plainer sailing and better water. But it would be insanity to attempt Hat Island at night. So there was a deal of looking at watches all the rest of the day, and a constant ciphering upon the speed we were making; Hat Island was the eternal subject; sometimes hope was high and sometimes we were delayed in a bad crossing, and down it went again. For hours all hands lay under the burden of this suppressed excitement; it was even communicated to me, and I got to feeling so solicitous about Hat Island, and under such an awful pressure of responsibility, that I wished I might have five minutes on shore to draw a good, full, relieving breath, and start over again. We were standing no regular watches. Each of our pilots ran such portions of the river as he had run when coming up-stream, because of his greater familiarity with it; but both remained in the pilot house constantly.

An hour before sunset, Mr. Bixby took the wheel and Mr. W——stepped aside. For the next thirty minutes every man held his watch in his hand and was restless, silent, and uneasy. At last somebody said, with a doomful sigh



'Well, yonder's Hat Island—and we can't make it.' All the watches closed with a snap, everybody sighed and muttered something about its being 'too bad, too bad—ah, if we could only have got here half an hour sooner!' and the place was thick with the atmosphere of disappointment. Some started to go out, but loitered, hearing no bell-tap to land. The sun dipped behind the horizon, the boat went on. Inquiring looks passed from one guest to another; and one who had his hand on the door-knob and had turned it, waited, then presently took away his hand and let the knob turn back again. We bore steadily down the bend. More looks were exchanged, and nods of surprised admiration—but no words. Insensibly the men drew together behind Mr. Bixby, as the sky darkened and one or two dim stars came out. The dead silence and sense of waiting became oppressive. Mr. Bixby pulled the cord, and two deep, mellow notes from the big bell floated off on the night. Then a pause, and one more note was struck. The watchman's voice followed, from the hurricane deck—

'Labboard lead, there! Stabboard lead!'

The cries of the leadsmen began to rise out of the distance, and were gruffly repeated by the word-passers on the hurricane deck.

'M-a-r-k three!.... M-a-r-k three!.... Quarter-less three! Half twain! Quarter twain! M-a-r-k twain! Quarter-less—'

Mr. Bixby pulled two bell-ropes, and was answered by faint jinglings far below in the engine room, and our speed slackened. The steam began to whistle through the gauge-cocks. The cries of the leadsmen went on—and it is a weird sound, always, in the night. Every pilot in the lot was watching now, with fixed eyes, and talking under his breath. Nobody was calm and easy but Mr. Bixby. He would put his wheel down and stand on a spoke, and as the steamer swung into her (to me) utterly invisible marks—for we seemed to be in the midst of a wide and gloomy sea—he would meet and fasten her there. Out of the murmur of half-audible talk, one caught a coherent sentence now and then—such as—

'There; she's over the first reef all right!'

After a pause, another subdued voice—

'Her stern's coming down just exactly right, by George!'

'Now she's in the marks; over she goes!'

Somebody else muttered—

'Oh, it was done beautiful—BEAUTIFUL!'

Now the engines were stopped altogether, and we drifted with the current. Not that I could see the boat drift, for I could not, the stars being all gone by this time. This drifting was the dismalest work; it held one's heart still. Presently I discovered a blacker gloom than that which surrounded us. It was the head of the island. We were closing right down upon it. We entered its deeper shadow, and so imminent seemed the peril that I was likely to suffocate; and I had the strongest impulse to do SOMETHING, anything, to save the vessel. But still Mr. Bixby stood by his wheel, silent, intent as a cat, and all the pilots stood shoulder to shoulder at his back.

'She'll not make it!' somebody whispered.

The water grew shoaler and shoaler, by the leadsman's cries, till it was down to—

'Eight-and-a-half!.... E-i-g-h-t feet!.... E-i-g-h-t feet!.... Seven-and—'

Mr. Bixby said warningly through his speaking tube to the engineer—

'Stand by, now!'

'Aye-aye, sir!'



'Seven-and-a-half! Seven feet! Six-and—'

We touched bottom! Instantly Mr. Bixby set a lot of bells ringing, shouted through the tube, 'NOW, let her have it—every ounce you've got!' then to his partner, 'Put her hard down! snatch her! snatch her!' The boat rasped and ground her way through the sand, hung upon the apex of disaster a single tremendous instant, and then over she went! And such a shout as went up at Mr. Bixby's back never loosened the roof of a pilot-house before!



"OVER SHE GOES."

There was no more trouble after that. Mr. Bixby was a hero that night; and it was some little time, too, before his exploit ceased to be talked about by river men.

Fully to realize the marvelous precision required in laying the great steamer in her marks in that murky waste of water, one should know that not only must she pick her intricate way through snags and blind reefs, and then shave the head of the island so closely as to brush the overhanging foliage with her stern, but at one place she must pass almost within arm's reach of a sunken and invisible wreck that would snatch the hull timbers from under her if she should strike it, and destroy a quarter of a million dollars' worth of steam-boat and cargo in five minutes, and maybe a hundred and fifty human lives into the bargain.

The last remark I heard that night was a compliment to Mr. Bixby, uttered in soliloquy and with unction by one of our guests. He said—

'By the Shadow of Death, but he's a lightning pilot!'



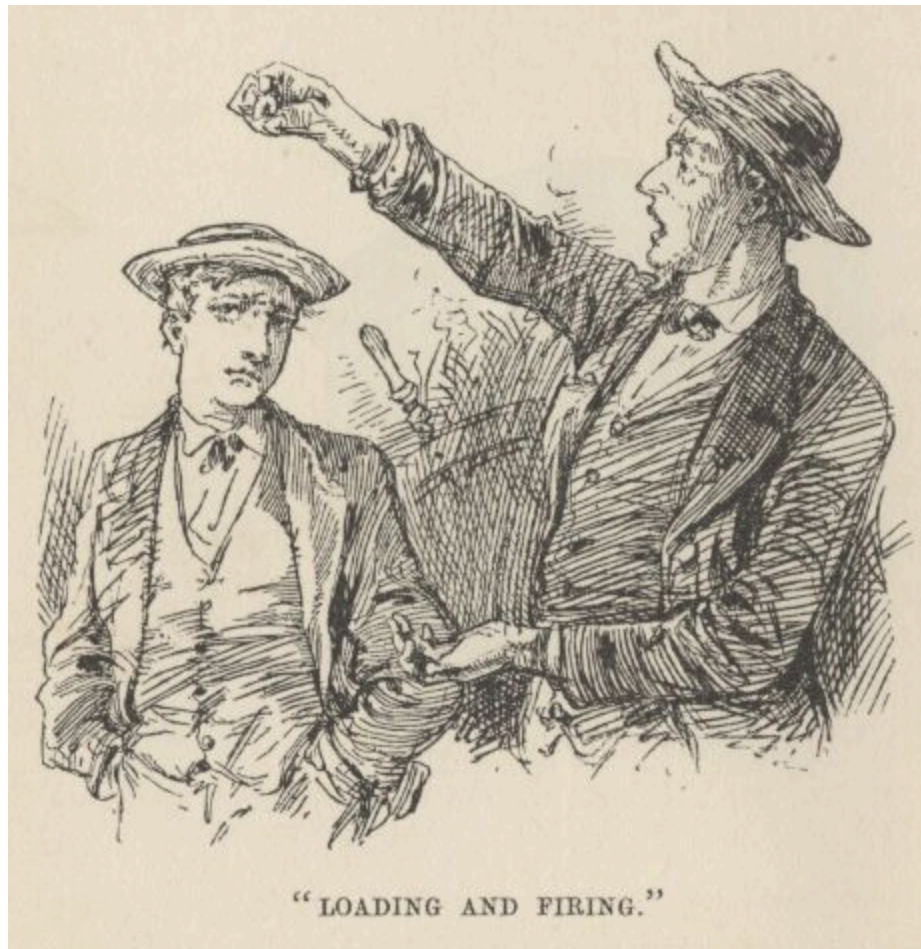
Chapter 8

Perplexing Lessons

At the end of what seemed a tedious while, I had managed to pack my head full of islands, towns, bars, 'points,' and bends; and a curiously inanimate mass of lumber it was, too. However, inasmuch as I could shut my eyes and reel off a good long string of these names without leaving out more than ten miles of river in every fifty, I began to feel that I could take a boat down to New Orleans if I could make her skip those little gaps. But of course my complacency could hardly get start enough to lift my nose a trifle into the air, before Mr. Bixby would think of something to fetch it down again. One day he turned on me suddenly with this settler—

'What is the shape of Walnut Bend?'

He might as well have asked me my grandmother's opinion of protoplasm. I reflected respectfully, and then said I didn't know it had any particular shape. My gunpowdery chief went off with a bang, of course, and then went on loading and firing until he was out of adjectives.



I had learned long ago that he only carried just so many rounds of ammunition, and was sure to subside into a very placable and even remorseful old smooth-bore as soon as they were all gone. That word 'old' is merely affectionate; he was not more than thirty-four. I waited. By and by he said—

'My boy, you've got to know the SHAPE of the river perfectly. It is all there is left to steer by on a very dark night. Everything else is blotted out and gone. But mind you, it hasn't the same shape in the night that it has in the day-time.'

'How on earth am I ever going to learn it, then?'

'How do you follow a hall at home in the dark. Because you know the shape of it. You can't see it.'

'Do you mean to say that I've got to know all the million trifling variations of shape in the banks of this interminable river as well as I know the shape of the front hall at home?'

'On my honor, you've got to know them BETTER than any man ever did know the shapes of the halls in his own house.'

'I wish I was dead!'

'Now I don't want to discourage you, but—'

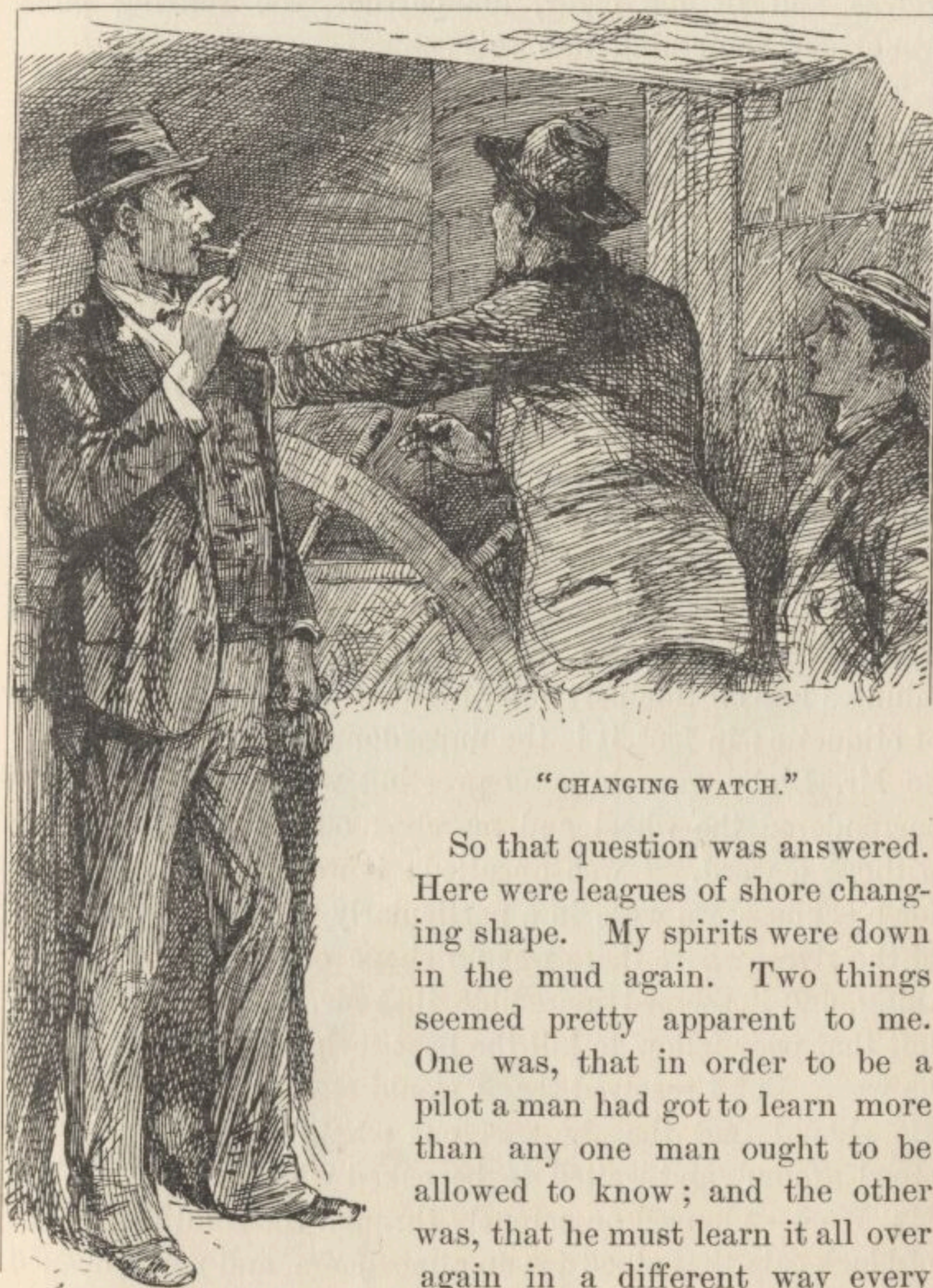
'Well, pile it on me; I might as well have it now as another time.'

'You see, this has got to be learned; there isn't any getting around it. A clear starlight night throws such heavy shadows that if you didn't know the shape of a shore perfectly you would claw away from every bunch of timber, because you would take the black shadow of it for a solid cape; and you see you would be getting scared to death every fifteen minutes by the watch. You would be fifty yards from shore all the time when you ought to be within fifty feet of it. You can't see a snag in one of those shadows, but you know exactly where it is, and the shape of the river tells you when you are coming to it. Then there's your pitch-dark night; the river is a very different shape on a pitch-dark night from what it is on a starlight night. All shores seem to be straight lines, then, and mighty dim ones, too; and you'd RUN them for straight lines only you know better. You boldly drive your boat right into what seems to be a solid, straight wall (you knowing very well that in reality there is a curve there), and that wall falls back and makes way for you. Then there's your gray mist. You take a night when there's one of these grisly, drizzly, gray mists, and then there isn't any particular shape to a shore. A gray mist would tangle the head of the oldest man that ever lived. Well, then, different kinds of MOONLIGHT change the shape of the river in different ways. You see—'

'Oh, don't say any more, please! Have I got to learn the shape of the river according to all these five hundred thousand different ways? If I tried to carry all that cargo in my head it would make me stoop-shouldered.'

'NO! you only learn THE shape of the river, and you learn it with such absolute certainty that you can always steer by the shape that's IN YOUR HEAD, and never mind the one that's before your eyes.'

'Very well, I'll try it; but after I have learned it can I depend on it. Will it keep the same form and not go fooling around?'



“CHANGING WATCH.”

So that question was answered. Here were leagues of shore changing shape. My spirits were down in the mud again. Two things seemed pretty apparent to me. One was, that in order to be a pilot a man had got to learn more than any one man ought to be allowed to know; and the other was, that he must learn it all over again in a different way every twenty-four hours.

Before Mr. Bixby could answer, Mr. W—— came in to take the watch, and he said—

'Bixby, you'll have to look out for President's Island and all that country clear away up above the Old Hen and Chickens. The banks are caving and the shape of the shores changing like everything. Why, you wouldn't know the point above 40. You can go up inside the old sycamore-s snag, now. {footnote [1. It may not be necessary, but still it can do no harm to explain that 'inside' means between the snag and the shore.—M.T.]}

So that question was answered. Here were leagues of shore changing shape. My spirits were down in the mud again. Two things seemed pretty apparent to me. One was, that in order to be a pilot a man had got to learn more than any one man ought to be allowed to know; and the other was, that he must learn it all over again in a different way every twenty-four hours.

That night we had the watch until twelve. Now it was an ancient river custom for the two pilots to chat a bit when the watch changed. While the relieving pilot put on his gloves and lit his cigar, his partner, the retiring pilot, would say something like this—

'I judge the upper bar is making down a little at Hale's Point; had quarter twain with the lower lead and mark twain {footnote [Two fathoms. 'Quarter twain' is two-and-a-quarter fathoms, thirteen-and-a-half feet. 'Mark three' is three fathoms.]} with the other.'

'Yes, I thought it was making down a little, last trip. Meet any boats?'

'Met one abreast the head of 21, but she was away over hugging the bar, and I couldn't make her out entirely. I took her for the "Sunny South"—hadn't any skylights forward of the chimneys.'

And so on. And as the relieving pilot took the wheel his partner {footnote ['Partner' is a technical term for 'the other pilot'.]} would mention that we were in such-and-such a bend, and say we were abreast of such-and-such a man's wood-yard or plantation. This was courtesy; I supposed it was necessity. But Mr. W—— came on watch full twelve minutes late on this

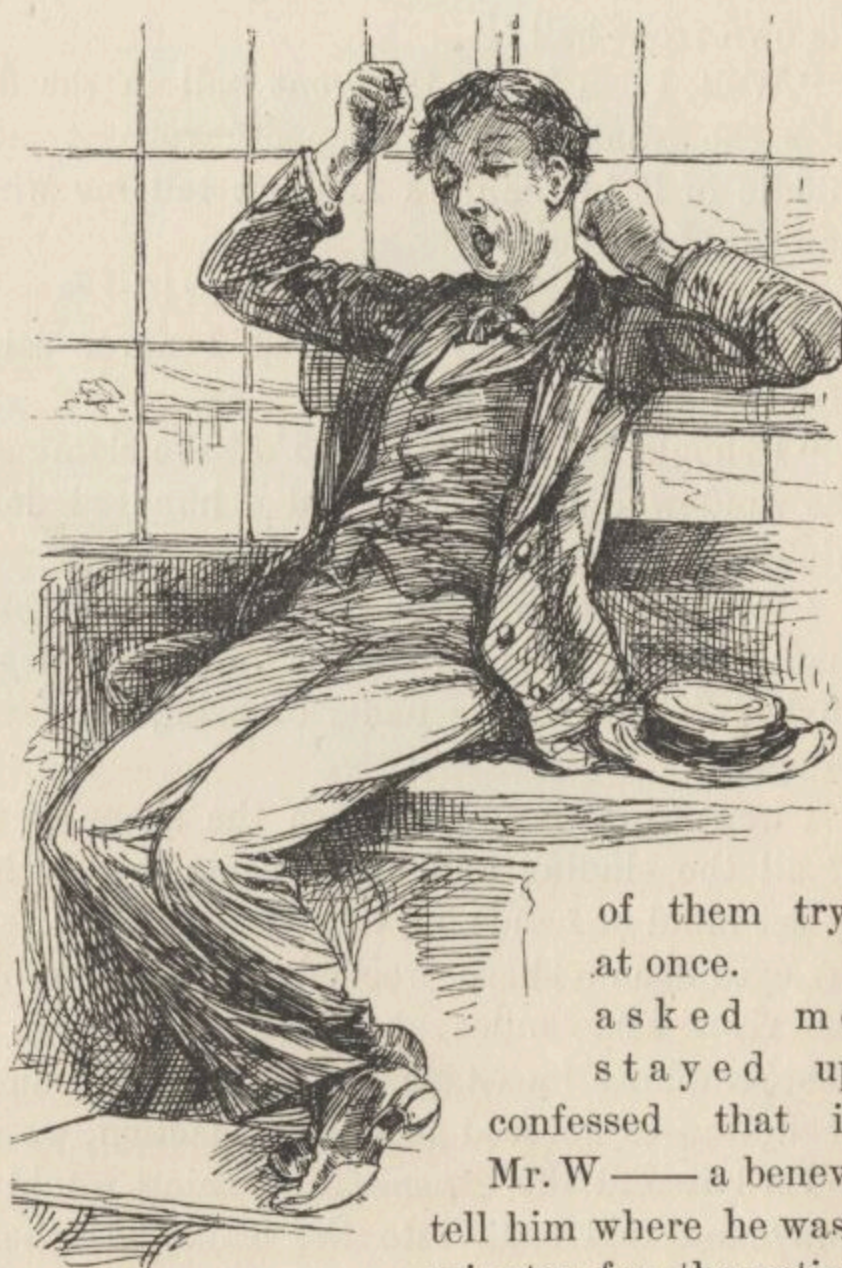
particular night,—a tremendous breach of etiquette; in fact, it is the unpardonable sin among pilots. So Mr. Bixby gave him no greeting whatever, but simply surrendered the wheel and marched out of the pilot-house without a word. I was appalled; it was a villainous night for blackness, we were in a particularly wide and blind part of the river, where there was no shape or substance to anything, and it seemed incredible that Mr. Bixby should have left that poor fellow to kill the boat trying to find out where he was. But I resolved that I would stand by him any way. He should find that he was not wholly friendless. So I stood around, and waited to be asked where we were. But Mr. W—— plunged on serenely through the solid firmament of black cats that stood for an atmosphere, and never opened his mouth. Here is a proud devil, thought I; here is a limb of Satan that would rather send us all to destruction than put himself under obligations to me, because I am not yet one of the salt of the earth and privileged to snub captains and lord it over everything dead and alive in a steamboat. I presently climbed up on the bench; I did not think it was safe to go to sleep while this lunatic was on watch.

However, I must have gone to sleep in the course of time, because the next thing I was aware of was the fact that day was breaking, Mr. W—— gone, and Mr. Bixby at the wheel again. So it was four o'clock and all well—but me; I felt like a skinful of dry bones and all of them trying to ache at once.

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"ALL WELL — BUT ME."

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Mr. Bixby asked me what I had stayed up there for. I confessed that it was to do Mr. W—— a benevolence,—tell him where he was. It took five minutes for the entire preposterousness of the thing to filter into Mr. Bixby's system, and then I judge it filled him nearly up to the chin; because he paid me a compliment—and not much of a one either. He said,

'Well, taking you by-and-large, you do seem to be more different kinds of an ass than any creature I ever saw before. What did you suppose he wanted to know for?'

I said I thought it might be a convenience to him.

'Convenience D-nation! Didn't I tell you that a man's got to know the river in the night the same as he'd know his own front hall?'

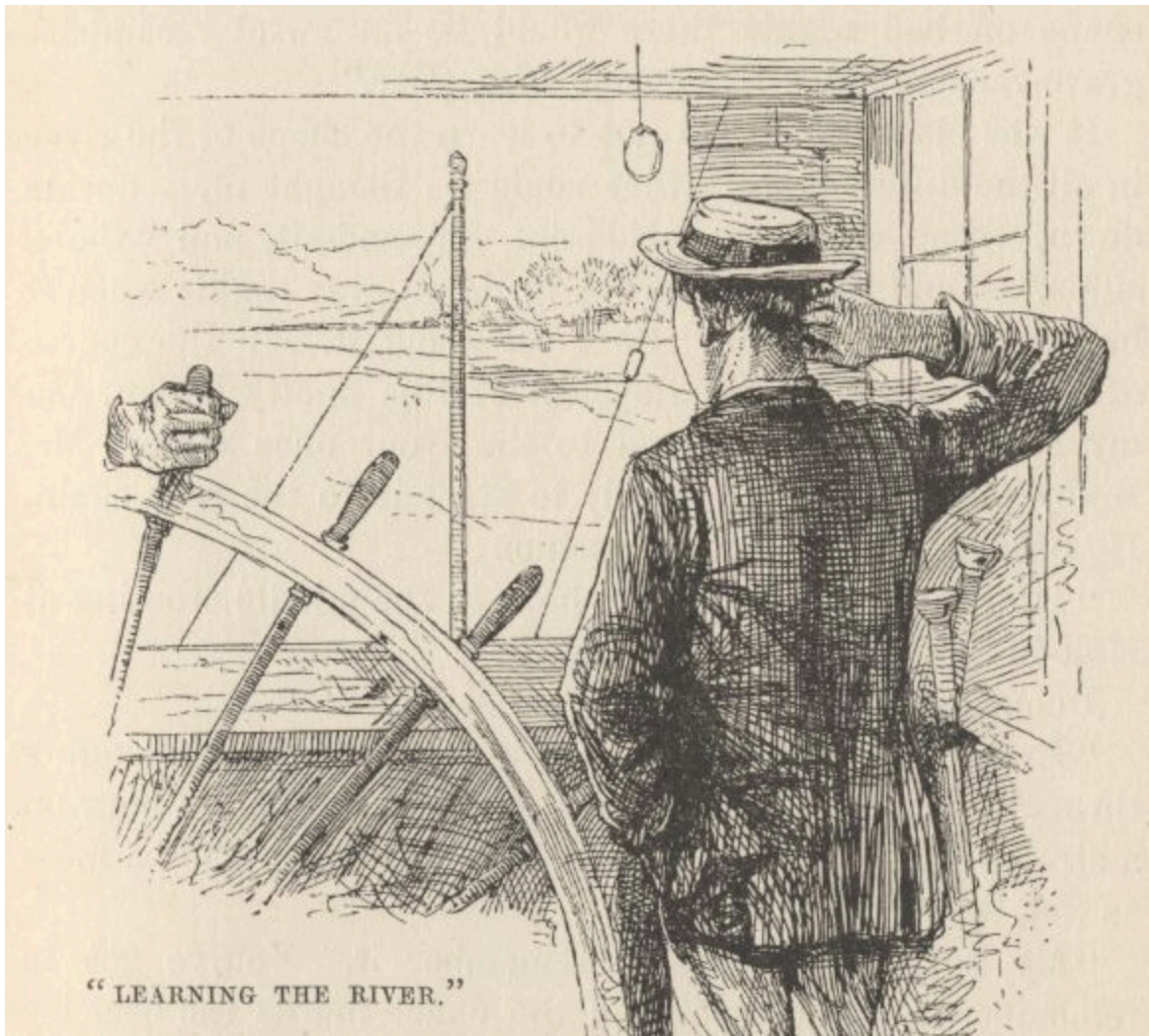
'Well, I can follow the front hall in the dark if I know it IS the front hall; but suppose you set me down in the middle of it in the dark and not tell me which hall it is; how am I to know?'

'Well you've GOT to, on the river!'

'All right. Then I'm glad I never said anything to Mr. W—— '

'I should say so. Why, he'd have slammed you through the window and utterly ruined a hundred dollars' worth of window-sash and stuff.'

I was glad this damage had been saved, for it would have made me unpopular with the owners. They always hated anybody who had the name of being careless, and injuring things.



I went to work now to learn the shape of the river; and of all the eluding and ungraspable objects that ever I tried to get mind or hands on, that was the chief. I would fasten my eyes upon a sharp, wooded point that projected far into the river some miles ahead of me, and go to laboriously photographing its shape upon my brain; and just as I was beginning to succeed to my satisfaction, we would draw up toward it and the exasperating thing would begin to melt away and fold back into the bank! If there had been a conspicuous dead tree standing upon the very point of the cape, I would find that tree inconspicuously merged into the general forest, and occupying the middle of a straight shore, when I got abreast of it! No prominent hill would stick to its shape long enough for me to make up my

mind what its form really was, but it was as dissolving and changeful as if it had been a mountain of butter in the hottest corner of the tropics. Nothing ever had the same shape when I was coming downstream that it had borne when I went up. I mentioned these little difficulties to Mr. Bixby. He said—

'That's the very main virtue of the thing. If the shapes didn't change every three seconds they wouldn't be of any use. Take this place where we are now, for instance. As long as that hill over yonder is only one hill, I can boom right along the way I'm going; but the moment it splits at the top and forms a V, I know I've got to scratch to starboard in a hurry, or I'll bang this boat's brains out against a rock; and then the moment one of the prongs of the V swings behind the other, I've got to waltz to larboard again, or I'll have a misunderstanding with a snag that would snatch the keelson out of this steamboat as neatly as if it were a sliver in your hand. If that hill didn't change its shape on bad nights there would be an awful steamboat graveyard around here inside of a year.'

It was plain that I had got to learn the shape of the river in all the different ways that could be thought of,—upside down, wrong end first, inside out, fore-and-aft, and 'thortships,'—and then know what to do on gray nights when it hadn't any shape at all. So I set about it. In the course of time I began to get the best of this knotty lesson, and my self-complacency moved to the front once more. Mr. Bixby was all fixed, and ready to start it to the rear again. He opened on me after this fashion—

'How much water did we have in the middle crossing at Hole-in-the-Wall, trip before last?'

I considered this an outrage. I said—

'Every trip, down and up, the leadsmen are singing through that tangled place for three-quarters of an hour on a stretch. How do you reckon I can remember such a mess as that?'

'My boy, you've got to remember it. You've got to remember the exact spot and the exact marks the boat lay in when we had the shoalest water, in everyone of the five hundred shoal places between St. Louis and New Orleans; and you mustn't get the shoal soundings and marks of one trip

mixed up with the shoal soundings and marks of another, either, for they're not often twice alike. You must keep them separate.'

When I came to myself again, I said—

'When I get so that I can do that, I'll be able to raise the dead, and then I won't have to pilot a steamboat to make a living. I want to retire from this business. I want a slush-bucket and a brush; I'm only fit for a roustabout. I haven't got brains enough to be a pilot; and if I had I wouldn't have strength enough to carry them around, unless I went on crutches.'

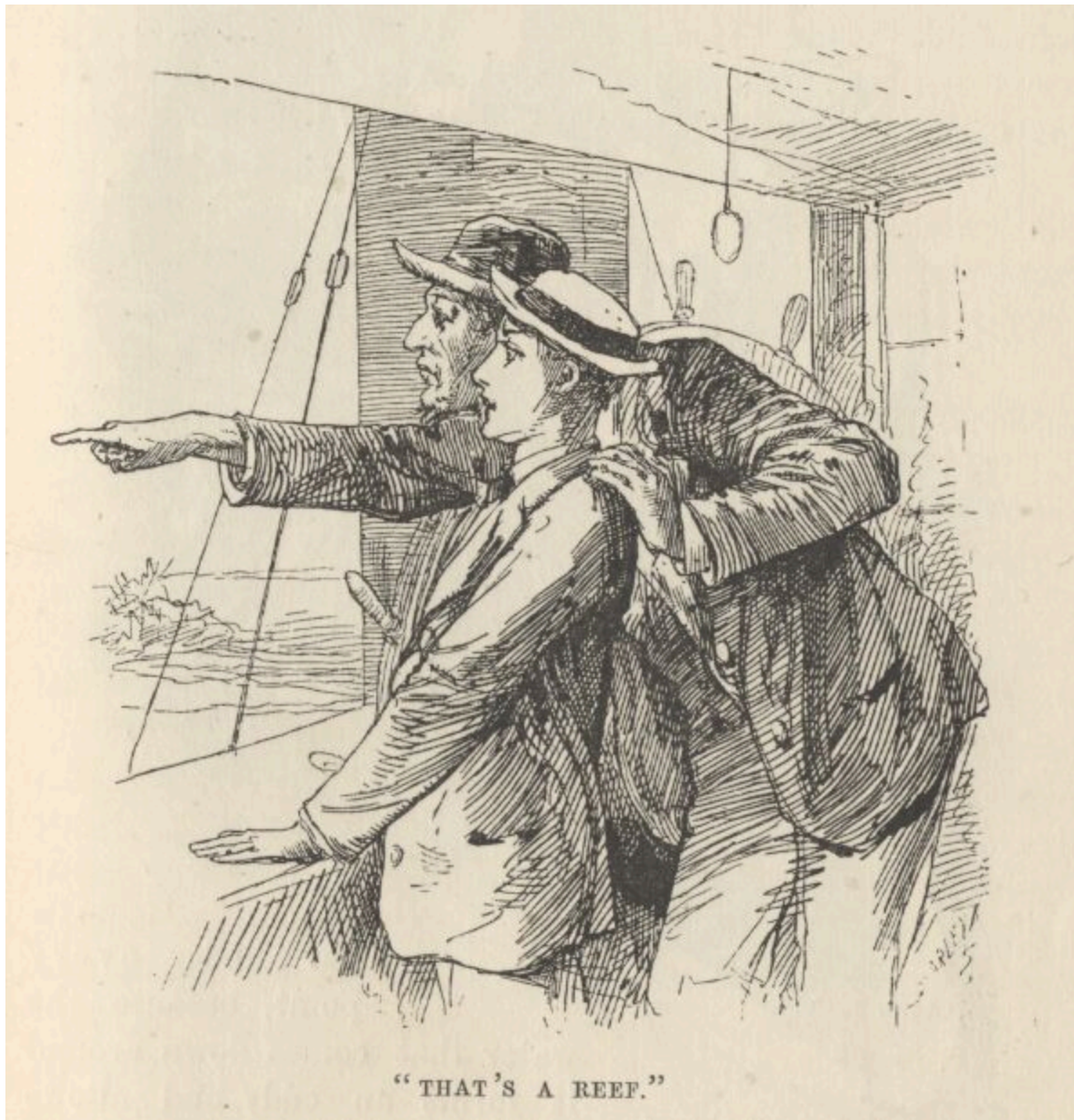
'Now drop that! When I say I'll learn {footnote ['Teach' is not in the river vocabulary.]} a man the river, I mean it. And you can depend on it, I'll learn him or kill him.'



Chapter 9

Continued Perplexities

THERE was no use in arguing with a person like this. I promptly put such a strain on my memory that by and by even the shoal water and the countless crossing-marks began to stay with me. But the result was just the same. I never could more than get one knotty thing learned before another presented itself. Now I had often seen pilots gazing at the water and pretending to read it as if it were a book; but it was a book that told me nothing. A time came at last, however, when Mr. Bixby seemed to think me far enough advanced to bear a lesson on water-reading. So he began—



'Do you see that long slanting line on the face of the water? Now, that's a reef. Moreover, it's a bluff reef. There is a solid sand-bar under it that is nearly as straight up and down as the side of a house. There is plenty of water close up to it, but mighty little on top of it. If you were to hit it you would knock the boat's brains out. Do you see where the line fringes out at the upper end and begins to fade away?'

'Yes, sir.'

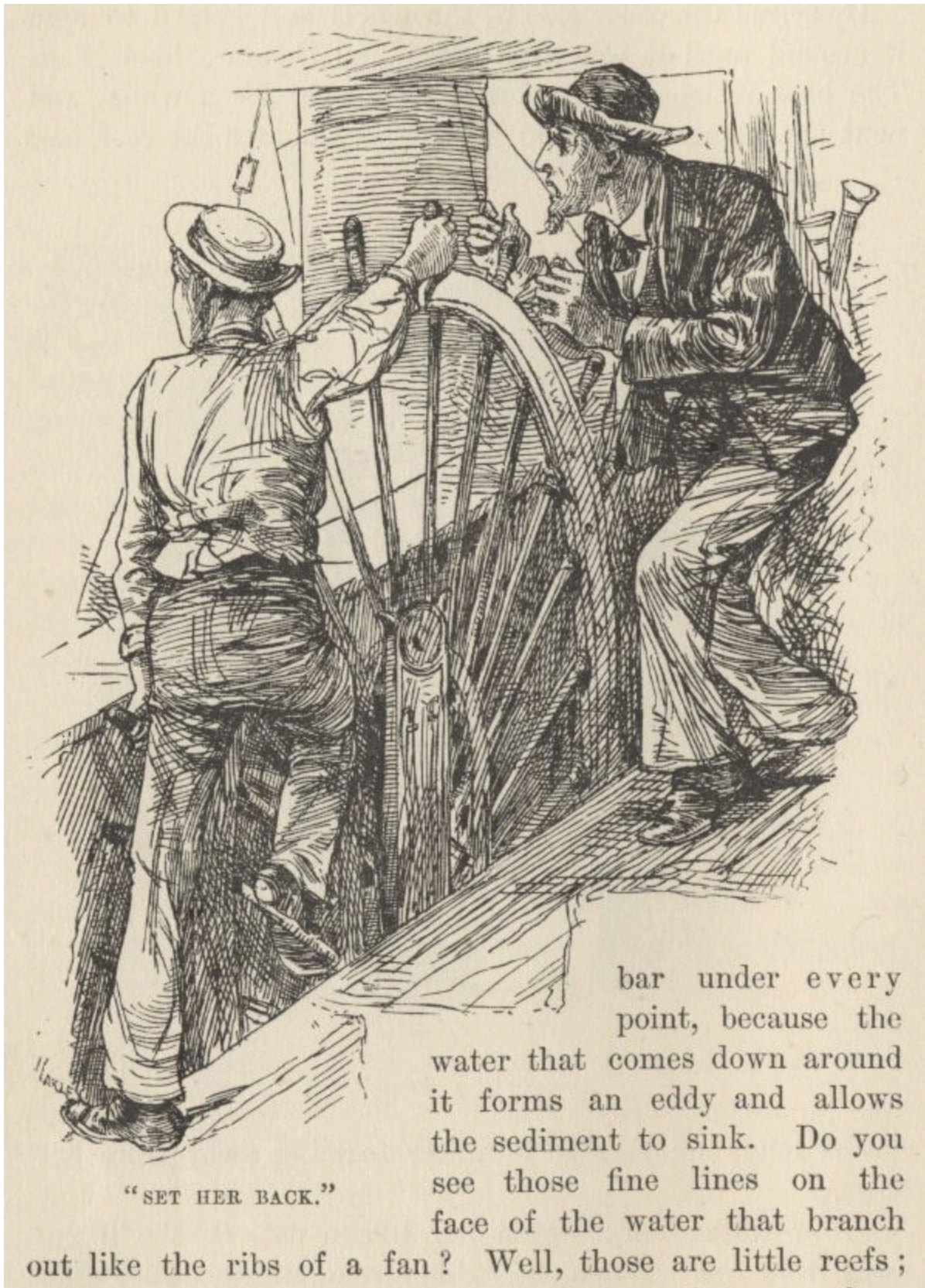
'Well, that is a low place; that is the head of the reef. You can climb over there, and not hurt anything. Cross over, now, and follow along close under the reef—easy water there—not much current.'

I followed the reef along till I approached the fringed end. Then Mr. Bixby said—

'Now get ready. Wait till I give the word. She won't want to mount the reef; a boat hates shoal water. Stand by—wait—WAIT—keep her well in hand. NOW cramp her down! Snatch her! snatch her!'

He seized the other side of the wheel and helped to spin it around until it was hard down, and then we held it so. The boat resisted, and refused to answer for a while, and next she came surging to starboard, mounted the reef, and sent a long, angry ridge of water foaming away from her bows.

'Now watch her; watch her like a cat, or she'll get away from you. When she fights strong and the tiller slips a little, in a jerky, greasy sort of way, let up on her a trifle; it is the way she tells you at night that the water is too shoal; but keep edging her up, little by little, toward the point. You are well up on the bar, now; there is a bar under every point, because the water that comes down around it forms an eddy and allows the sediment to sink. Do you see those fine lines on the face of the water that branch out like the ribs of a fan. Well, those are little reefs; you want to just miss the ends of them, but run them pretty close. Now look out—look out! Don't you crowd that slick, greasy-looking place; there ain't nine feet there; she won't stand it. She begins to smell it; look sharp, I tell you! Oh blazes, there you go! Stop the starboard wheel! Quick! Ship up to back! Set her back!



"SET HER BACK."

bar under every point, because the water that comes down around it forms an eddy and allows the sediment to sink. Do you see those fine lines on the face of the water that branch out like the ribs of a fan? Well, those are little reefs;

The engine bells jingled and the engines answered promptly, shooting white columns of steam far aloft out of the 'scape pipes, but it was too late. The boat had 'smelt' the bar in good earnest; the foamy ridges that radiated from her bows suddenly disappeared, a great dead swell came rolling forward and swept ahead of her, she careened far over to larboard, and went tearing away toward the other shore as if she were about scared to death. We were a good mile from where we ought to have been, when we finally got the upper hand of her again.

During the afternoon watch the next day, Mr. Bixby asked me if I knew how to run the next few miles. I said—

'Go inside the first snag above the point, outside the next one, start out from the lower end of Higgins's wood-yard, make a square crossing and—'

'That's all right. I'll be back before you close up on the next point.'

But he wasn't. He was still below when I rounded it and entered upon a piece of river which I had some misgivings about. I did not know that he was hiding behind a chimney to see how I would perform. I went gaily along, getting prouder and prouder, for he had never left the boat in my sole charge such a length of time before. I even got to 'setting' her and letting the wheel go, entirely, while I vaingloriously turned my back and inspected the stem marks and hummed a tune, a sort of easy indifference which I had prodigiously admired in Bixby and other great pilots. Once I inspected rather long, and when I faced to the front again my heart flew into my mouth so suddenly that if I hadn't clapped my teeth together I should have lost it. One of those frightful bluff reefs was stretching its deadly length right across our bows! My head was gone in a moment; I did not know which end I stood on; I gasped and could not get my breath; I spun the wheel down with such rapidity that it wove itself together like a spider's web; the boat answered and turned square away from the reef, but the reef followed her! I fled, and still it followed, still it kept—right across my bows! I never looked to see where I was going, I only fled. The awful crash was imminent—why didn't that villain come! If I committed the crime of ringing a bell, I might get thrown overboard. But better that than kill the boat. So in blind desperation I started such a rattling 'shivaree' down below

as never had astounded an engineer in this world before, I fancy. Amidst the frenzy of the bells the engines began to back and fill in a furious way, and my reason forsook its throne—we were about to crash into the woods on the other side of the river. Just then Mr. Bixby stepped calmly into view on the hurricane deck. My soul went out to him in gratitude. My distress vanished; I would have felt safe on the brink of Niagara, with Mr. Bixby on the hurricane deck. He blandly and sweetly took his tooth-pick out of his mouth between his fingers, as if it were a cigar—we were just in the act of climbing an overhanging big tree, and the passengers were scudding astern like rats—and lifted up these commands to me ever so gently—

'Stop the starboard. Stop the larboard. Set her back on both.'

The boat hesitated, halted, pressed her nose among the boughs a critical instant, then reluctantly began to back away.

'Stop the larboard. Come ahead on it. Stop the starboard. Come ahead on it. Point her for the bar.'

I blushed under the sarcasm, and said I had n't had any hail.

"Ah! Then it was for wood, I suppose. The officer of the watch will tell you when he wants to wood up."

I went on consuming, and said I was n't after wood.

"Indeed? Why, what could you want over here in the bend, then? Did you ever know of a boat following a bend up-stream at this stage of the river?"

"No, sir, — and *I* was n't trying to follow it. I was getting away from a bluff reef."

"No, it was n't a bluff reef; there is n't one within three miles of where you were."

"But I saw it. It was as bluff as that one yonder."

"Just about. Run over it!"

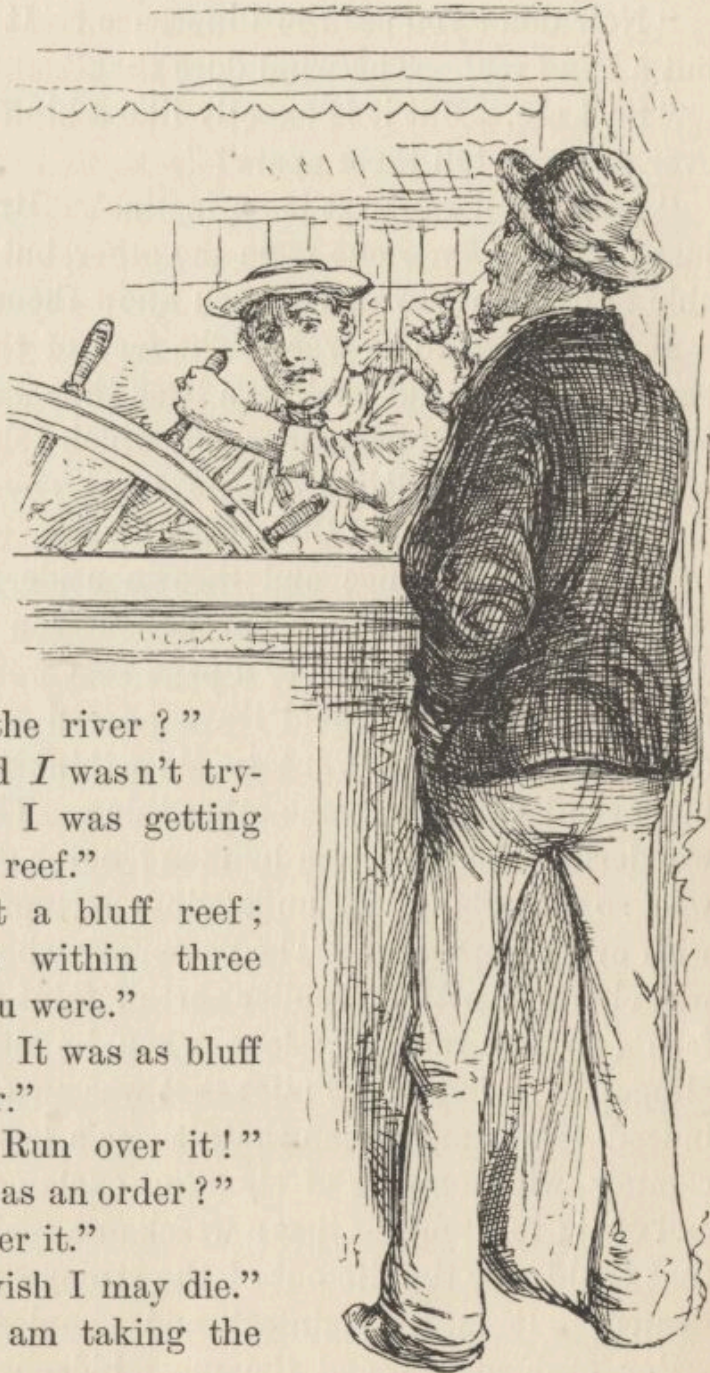
"Do you give it as an order?"

"Yes. Run over it."

"If I don't, I wish I may die."

"All right; I am taking the responsibility."

I was just as anxious to kill the boat, now, as I had been to save her before. I impressed my orders upon my memory, to be



MR. B. STEPPED INTO VIEW.

I sailed away as serenely as a summer's morning Mr. Bixby came in and said, with mock simplicity—

'When you have a hail, my boy, you ought to tap the big bell three times before you land, so that the engineers can get ready.'

I blushed under the sarcasm, and said I hadn't had any hail.

'Ah! Then it was for wood, I suppose. The officer of the watch will tell you when he wants to wood up.'

I went on consuming and said I wasn't after wood.

'Indeed? Why, what could you want over here in the bend, then? Did you ever know of a boat following a bend up-stream at this stage of the river?'

'No sir,—and I wasn't trying to follow it. I was getting away from a bluff reef.'

'No, it wasn't a bluff reef; there isn't one within three miles of where you were.'

'But I saw it. It was as bluff as that one yonder.'

'Just about. Run over it!'

'Do you give it as an order?'

'Yes. Run over it.'

'If I don't, I wish I may die.'

'All right; I am taking the responsibility.' I was just as anxious to kill the boat, now, as I had been to save her before. I impressed my orders upon my memory, to be used at the inquest, and made a straight break for the reef. As it disappeared under our bows I held my breath; but we slid over it like oil.

'Now don't you see the difference? It wasn't anything but a WIND reef. The wind does that.'

'So I see. But it is exactly like a bluff reef. How am I ever going to tell them apart?'

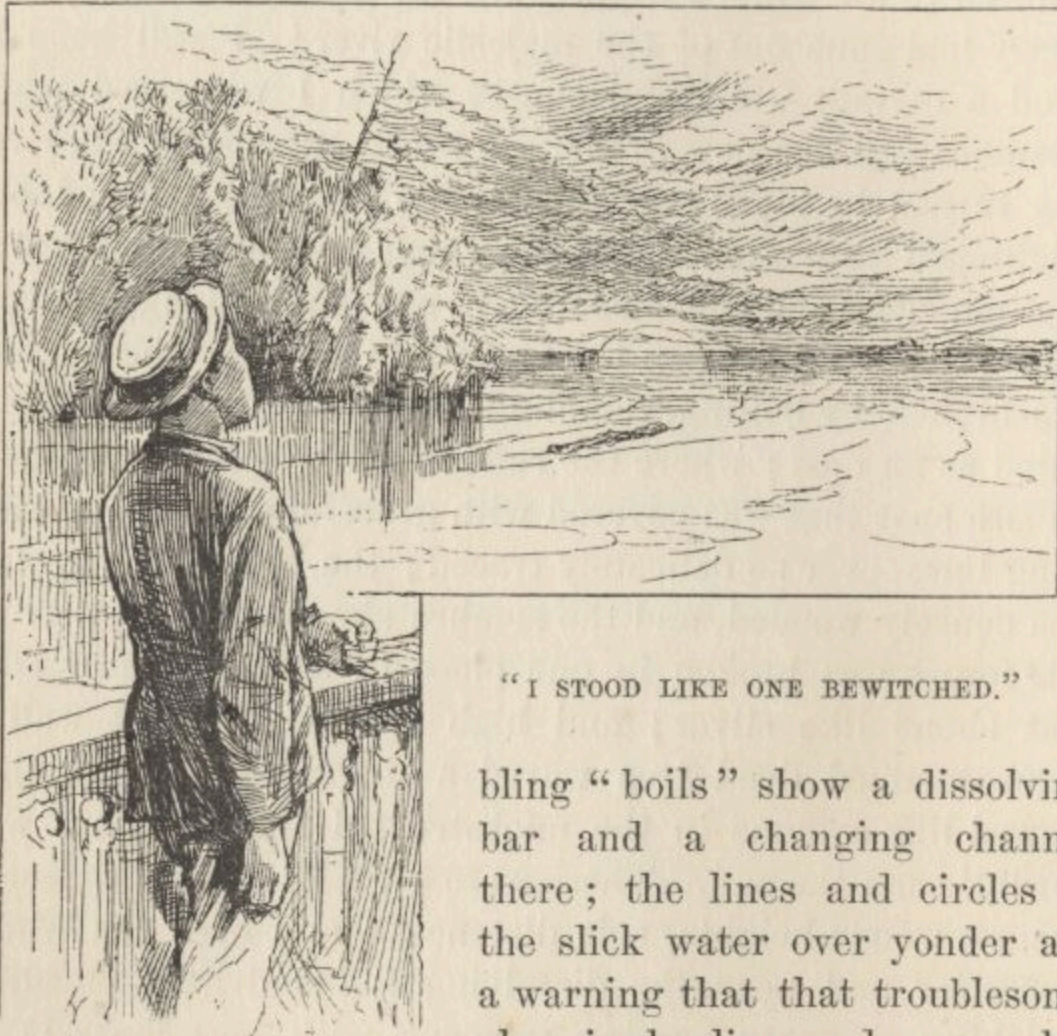
'I can't tell you. It is an instinct. By and by you will just naturally KNOW one from the other, but you never will be able to explain why or how you know them apart'

It turned out to be true. The face of the water, in time, became a wonderful book—a book that was a dead language to the uneducated passenger, but which told its mind to me without reserve, delivering its most cherished secrets as clearly as if it uttered them with a voice. And it was not a book to be read once and thrown aside, for it had a new story to tell every day. Throughout the long twelve hundred miles there was never a page that was void of interest, never one that you could leave unread without loss, never one that you would want to skip, thinking you could find higher enjoyment in some other thing. There never was so wonderful a book written by man; never one whose interest was so absorbing, so unflagging, so sparkingly renewed with every reperusal. The passenger who could not read it was charmed with a peculiar sort of faint dimple on its surface (on the rare occasions when he did not overlook it altogether); but to the pilot that was an *ITALICIZED* passage; indeed, it was more than that, it was a legend of the largest capitals, with a string of shouting exclamation points at the end of it; for it meant that a wreck or a rock was buried there that could tear the life out of the strongest vessel that ever floated. It is the faintest and simplest expression the water ever makes, and the most hideous to a pilot's eye. In truth, the passenger who could not read this book saw nothing but all manner of pretty pictures in it painted by the sun and shaded by the clouds, whereas to the trained eye these were not pictures at all, but the grimmest and most dead-earnest of reading-matter.

Now when I had mastered the language of this water and had come to know every trifling feature that bordered the great river as familiarly as I knew the letters of the alphabet, I had made a valuable acquisition. But I had lost something, too. I had lost something which could never be restored to me while I lived. All the grace, the beauty, the poetry had gone out of the majestic river! I still keep in mind a certain wonderful sunset which I witnessed when steamboating was new to me. A broad expanse of the river was turned to blood; in the middle distance the red hue brightened into

gold, through which a solitary log came floating, black and conspicuous; in one place a long, slanting mark lay sparkling upon the water; in another the surface was broken by boiling, tumbling rings, that were as many-tinted as an opal; where the ruddy flush was faintest, was a smooth spot that was covered with graceful circles and radiating lines, ever so delicately traced; the shore on our left was densely wooded, and the somber shadow that fell from this forest was broken in one place by a long, ruffled trail that shone like silver; and high above the forest wall a clean-stemmed dead tree waved a single leafy bough that glowed like a flame in the unobstructed splendor that was flowing from the sun. There were graceful curves, reflected images, woody heights, soft distances; and over the whole scene, far and near, the dissolving lights drifted steadily, enriching it, every passing moment, with new marvels of coloring.

This sun means that we are going to have wind to-morrow ; that floating log means that the river is rising, small thanks to it ; that slanting mark on the water refers to a bluff reef which is going to kill somebody's steamboat one of these nights, if it keeps on stretching out like that ; those tum-



"I STOOD LIKE ONE BEWITCHED."

bling "boils" show a dissolving bar and a changing channel there ; the lines and circles in the slick water over yonder are a warning that that troublesome place is shoaling up dangerously ;

I stood like one bewitched. I drank it in, in a speechless rapture. The world was new to me, and I had never seen anything like this at home. But as I have said, a day came when I began to cease from noting the glories and the charms which the moon and the sun and the twilight wrought upon

the river's face; another day came when I ceased altogether to note them. Then, if that sunset scene had been repeated, I should have looked upon it without rapture, and should have commented upon it, inwardly, after this fashion: This sun means that we are going to have wind to-morrow; that floating log means that the river is rising, small thanks to it; that slanting mark on the water refers to a bluff reef which is going to kill somebody's steamboat one of these nights, if it keeps on stretching out like that; those tumbling 'boils' show a dissolving bar and a changing channel there; the lines and circles in the slick water over yonder are a warning that that troublesome place is shoaling up dangerously; that silver streak in the shadow of the forest is the 'break' from a new snag, and he has located himself in the very best place he could have found to fish for steamboats; that tall dead tree, with a single living branch, is not going to last long, and then how is a body ever going to get through this blind place at night without the friendly old landmark.

No, the romance and the beauty were all gone from the river. All the value any feature of it had for me now was the amount of usefulness it could furnish toward compassing the safe piloting of a steamboat. Since those days, I have pitied doctors from my heart. What does the lovely flush in a beauty's cheek mean to a doctor but a 'break' that ripples above some deadly disease. Are not all her visible charms sown thick with what are to him the signs and symbols of hidden decay? Does he ever see her beauty at all, or doesn't he simply view her professionally, and comment upon her unwholesome condition all to himself? And doesn't he sometimes wonder whether he has gained most or lost most by learning his trade?



Chapter 10

Completing My Education

WHOSOEVER has done me the courtesy to read my chapters which have preceded this may possibly wonder that I deal so minutely with piloting as a science. It was the prime purpose of those chapters; and I am not quite done yet. I wish to show, in the most patient and painstaking way, what a wonderful science it is. Ship channels are buoyed and lighted, and therefore it is a comparatively easy undertaking to learn to run them; clear-water rivers, with gravel bottoms, change their channels very gradually, and therefore one needs to learn them but once; but piloting becomes another matter when you apply it to vast streams like the Mississippi and the Missouri, whose alluvial banks cave and change constantly, whose snags are always hunting up new quarters, whose sandbars are never at rest, whose channels are for ever dodging and shirking, and whose obstructions must be confronted in all nights and all weathers without the aid of a single light-house or a single buoy; for there is neither light nor buoy to be found anywhere in all this three or four thousand miles of villainous river. {footnote [True at the time referred to; not true now (1882).]} I feel justified in enlarging upon this great science for the reason that I feel sure no one has ever yet written a paragraph about it who had piloted a steamboat himself, and so had a practical knowledge of the subject. If the theme were hackneyed, I should be obliged to deal gently with the reader; but since it is wholly new, I have felt at liberty to take up a considerable degree of room with it.

When I had learned the name and position of every visible feature of the river; when I had so mastered its shape that I could shut my eyes and trace it from St. Louis to New Orleans; when I had learned to read the face of the water as one would cull the news from the morning paper; and finally, when I had trained my dull memory to treasure up an endless array of soundings

and crossing-marks, and keep fast hold of them, I judged that my education was complete: so I got to tilting my cap to the side of my head, and wearing a tooth-pick in my mouth at the wheel. Mr. Bixby had his eye on these airs. One day he said—



'What is the height of that bank yonder, at Burgess's?'

'How can I tell, sir. It is three-quarters of a mile away.'

'Very poor eye—very poor. Take the glass.'

I took the glass, and presently said—'I can't tell. I suppose that that bank is about a foot and a half high.'

'Foot and a half! That's a six-foot bank. How high was the bank along here last trip?'

'I don't know; I never noticed.'

'You didn't? Well, you must always do it hereafter.'

'Why?'

'Because you'll have to know a good many things that it tells you. For one thing, it tells you the stage of the river—tells you whether there's more water or less in the river along here than there was last trip.'

'The leads tell me that.' I rather thought I had the advantage of him there.

'Yes, but suppose the leads lie? The bank would tell you so, and then you'd stir those leadsmen up a bit. There was a ten-foot bank here last trip, and there is only a six-foot bank now. What does that signify?'

'That the river is four feet higher than it was last trip.'

'Very good. Is the river rising or falling?'

'Rising.'

'No it ain't.'

'I guess I am right, sir. Yonder is some drift-wood floating down the stream.'

'A rise starts the drift-wood, but then it keeps on floating a while after the river is done rising. Now the bank will tell you about this. Wait till you come to a place where it shelves a little. Now here; do you see this narrow belt of fine sediment That was deposited while the water was higher. You see the driftwood begins to strand, too. The bank helps in other ways. Do you see that stump on the false point?'



'Ay, ay, sir.'

'Well, the water is just up to the roots of it. You must make a note of that.'

'Why?'

'Because that means that there's seven feet in the chute of 103.'

'But 103 is a long way up the river yet.'

'That's where the benefit of the bank comes in. There is water enough in 103 NOW, yet there may not be by the time we get there; but the bank will keep us posted all along. You don't run close chutes on a falling river, upstream, and there are precious few of them that you are allowed to run at all

down-stream. There's a law of the United States against it. The river may be rising by the time we get to 103, and in that case we'll run it. We are drawing—how much?

'Six feet aft,—six and a half forward.'

'Well, you do seem to know something.'

'But what I particularly want to know is, if I have got to keep up an everlasting measuring of the banks of this river, twelve hundred miles, month in and month out?'

'Of course!'

My emotions were too deep for words for a while. Presently I said—'

And how about these chutes. Are there many of them?'

'I should say so. I fancy we shan't run any of the river this trip as you've ever seen it run before—so to speak. If the river begins to rise again, we'll go up behind bars that you've always seen standing out of the river, high and dry like the roof of a house; we'll cut across low places that you've never noticed at all, right through the middle of bars that cover three hundred acres of river; we'll creep through cracks where you've always thought was solid land; we'll dart through the woods and leave twenty-five miles of river off to one side; we'll see the hind-side of every island between New Orleans and Cairo.'

'Then I've got to go to work and learn just as much more river as I already know.'

'Just about twice as much more, as near as you can come at it.'

'Well, one lives to find out. I think I was a fool when I went into this business.'

'Yes, that is true. And you are yet. But you'll not be when you've learned it.'

'Ah, I never can learn it.'

'I will see that you DO.'

By and by I ventured again—

'Have I got to learn all this thing just as I know the rest of the river—shapes and all—and so I can run it at night?'

'Yes. And you've got to have good fair marks from one end of the river to the other, that will help the bank tell you when there is water enough in each of these countless places—like that stump, you know. When the river first begins to rise, you can run half a dozen of the deepest of them; when it rises a foot more you can run another dozen; the next foot will add a couple of dozen, and so on: so you see you have to know your banks and marks to a dead moral certainty, and never get them mixed; for when you start through one of those cracks, there's no backing out again, as there is in the big river; you've got to go through, or stay there six months if you get caught on a falling river. There are about fifty of these cracks which you can't run at all except when the river is brim full and over the banks.'

'This new lesson is a cheerful prospect.'

'Cheerful enough. And mind what I've just told you; when you start into one of those places you've got to go through. They are too narrow to turn around in, too crooked to back out of, and the shoal water is always up at the head; never elsewhere. And the head of them is always likely to be filling up, little by little, so that the marks you reckon their depth by, this season, may not answer for next.'

'Learn a new set, then, every year?'

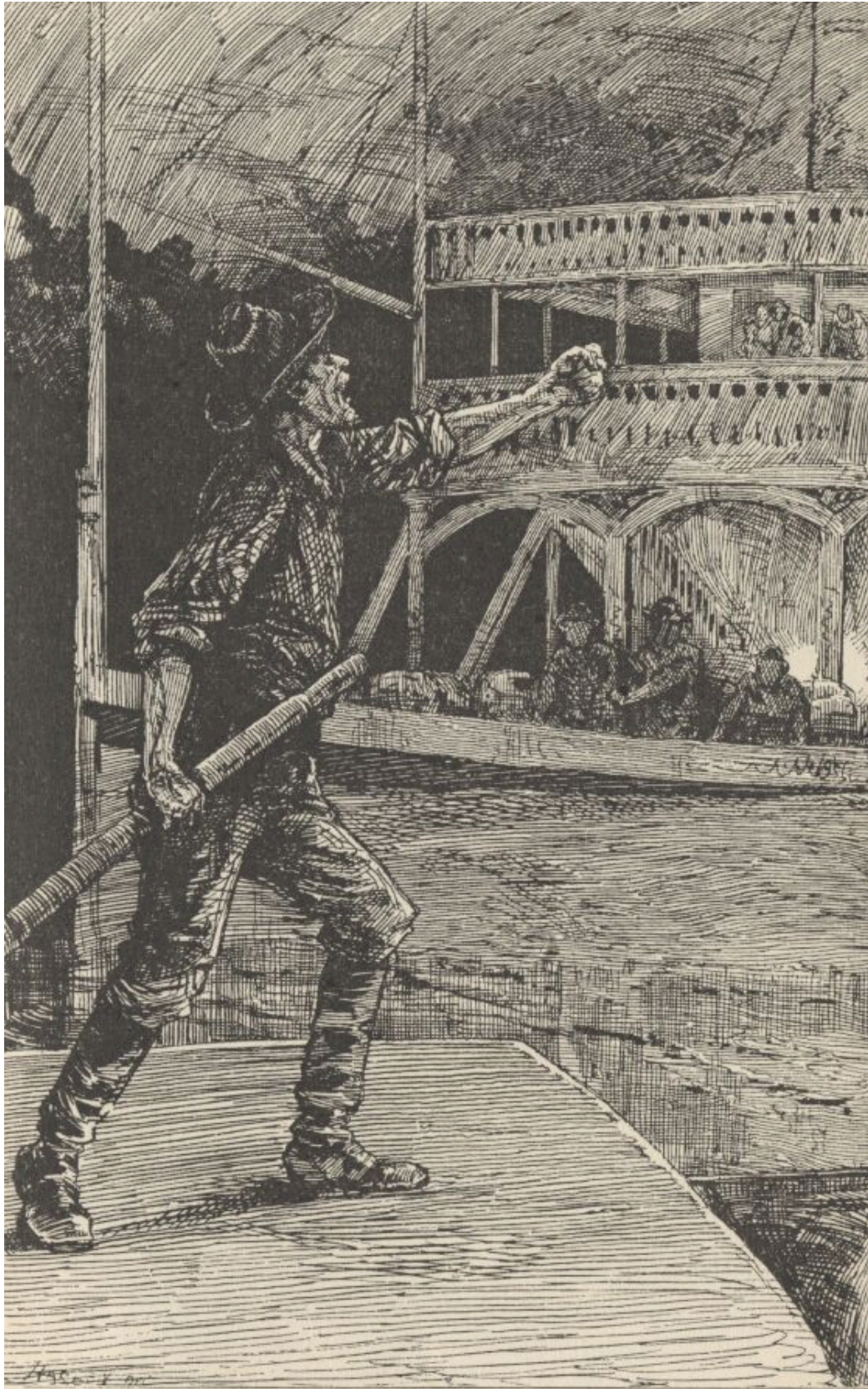
'Exactly. Cramp her up to the bar! What are you standing up through the middle of the river for?'

The next few months showed me strange things. On the same day that we held the conversation above narrated, we met a great rise coming down the river. The whole vast face of the stream was black with drifting dead logs, broken boughs, and great trees that had caved in and been washed away. It required the nicest steering to pick one's way through this rushing raft, even in the day-time, when crossing from point to point; and at night the difficulty was mightily increased; every now and then a huge log, lying deep in the water, would suddenly appear right under our bows, coming head-on; no use to try to avoid it then; we could only stop the engines, and

one wheel would walk over that log from one end to the other, keeping up a thundering racket and careening the boat in a way that was very uncomfortable to passengers. Now and then we would hit one of these sunken logs a rattling bang, dead in the center, with a full head of steam, and it would stun the boat as if she had hit a continent. Sometimes this log would lodge, and stay right across our nose, and back the Mississippi up before it; we would have to do a little craw-fishing, then, to get away from the obstruction. We often hit WHITE logs, in the dark, for we could not see them till we were right on them; but a black log is a pretty distinct object at night. A white snag is an ugly customer when the daylight is gone.

Of course, on the great rise, down came a swarm of prodigious timber-rafts from the head waters of the Mississippi, coal barges from Pittsburgh, little trading scows from everywhere, and broad-horns from 'Posey County,' Indiana, freighted with 'fruit and furniture'—the usual term for describing it, though in plain English the freight thus aggrandized was hoop-poles and pumpkins. Pilots bore a mortal hatred to these craft; and it was returned with usury. The law required all such helpless traders to keep a light burning, but it was a law that was often broken. All of a sudden, on a murky night, a light would hop up, right under our bows, almost, and an agonized voice, with the backwoods 'whang' to it, would wail out—

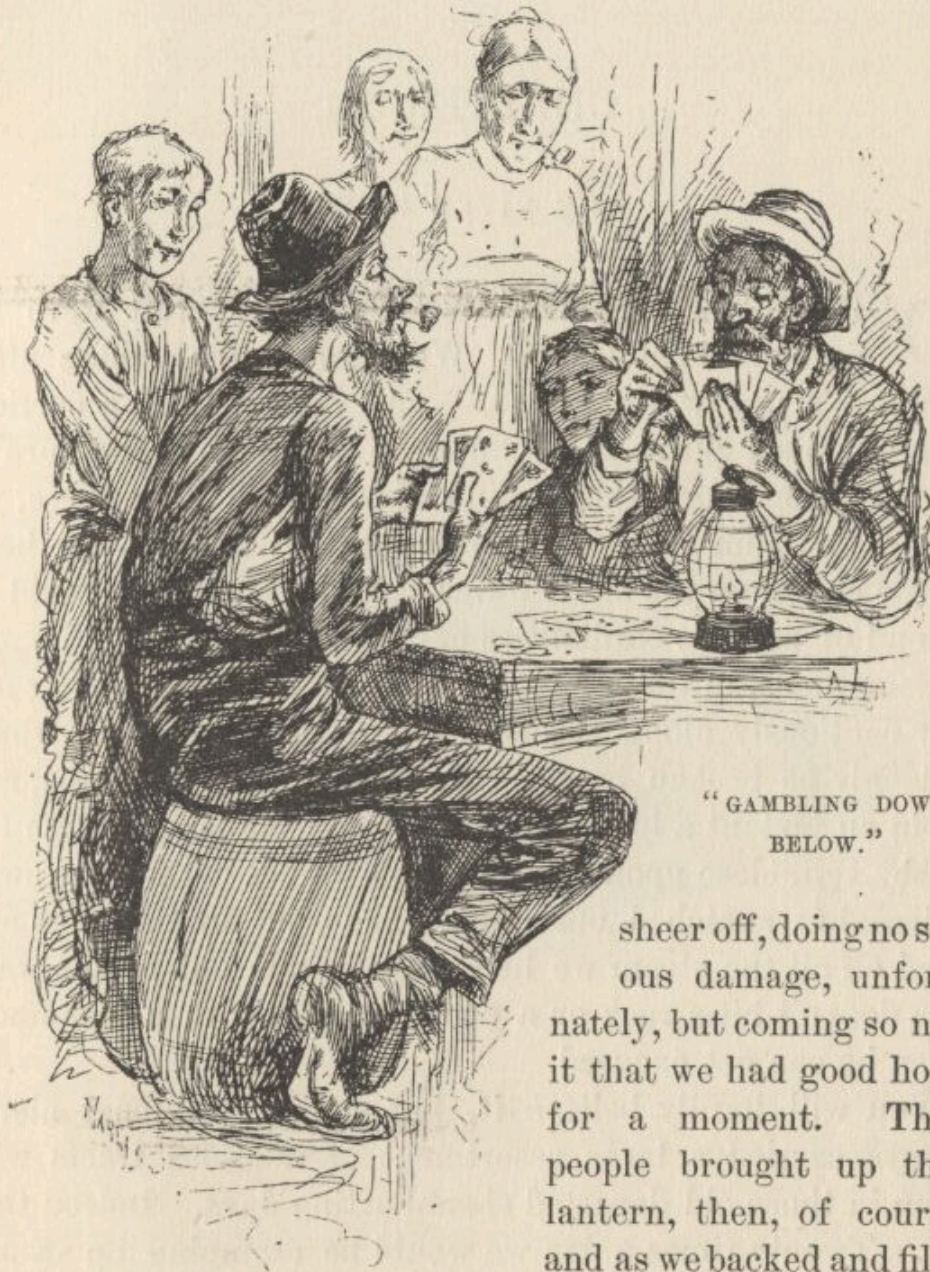
'Whar'n the —— you goin' to! Cain't you see nothin', you dash-dashed aig-suckin', sheep-stealin', one-eyed son of a stuffed monkey!'



THE ORATOR OF THE SCOW.

Then for an instant, as we whistled by, the red glare from our furnaces would reveal the scow and the form of the gesticulating orator as if under a lightning-flash, and in that instant our firemen and deck-hands would send and receive a tempest of missiles and profanity, one of our wheels would walk off with the crashing fragments of a steering-oar, and down the dead blackness would shut again. And that flatboatman would be sure to go into New Orleans and sue our boat, swearing stoutly that he had a light burning all the time, when in truth his gang had the lantern down below to sing and lie and drink and gamble by, and no watch on deck.

we should have eaten up a Posey County family, fruit, furniture, and all, but that they happened to be fiddling down below and we just caught the sound of the music in time to



"GAMBLING DOWN
BELOW."

sheer off, doing no serious damage, unfortunately, but coming so near it that we had good hopes for a moment. These people brought up their lantern, then, of course; and as we backed and filled to get away, the precious

family stood in the light of it—both sexes and various ages—and cursed us till everything turned blue. Once a coal-boatman sent a bullet through our pilot-house, when we borrowed a steering-oar of him in a very narrow place.

Once, at night, in one of those forest-bordered crevices (behind an island) which steamboatmen intensely describe with the phrase 'as dark as the inside of a cow,' we should have eaten up a Posey County family, fruit, furniture, and all, but that they happened to be fiddling down below, and we just caught the sound of the music in time to sheer off, doing no serious damage, unfortunately, but coming so near it that we had good hopes for a moment. These people brought up their lantern, then, of course; and as we backed and filled to get away, the precious family stood in the light of it—both sexes and various ages—and cursed us till everything turned blue. Once a coalboatman sent a bullet through our pilot-house, when we borrowed a steering oar of him in a very narrow place.

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